Digital Image Processing 9:

Color Image Processing

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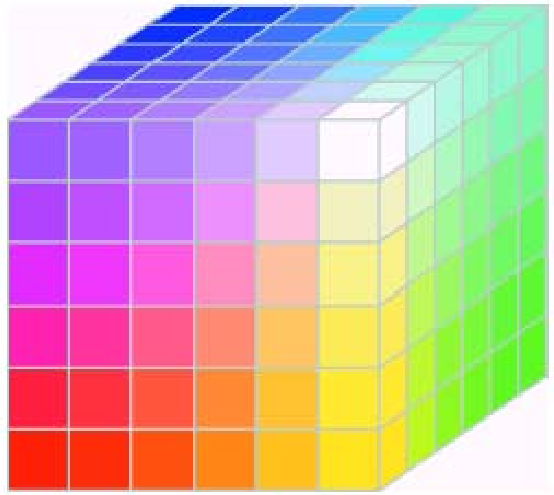
Today we'll look at color image processing, covering:

- Color fundamentals
- CIE Chromacity Diagram
- Color models
- Pseudocolor Image Processing
- Full-color Image processing



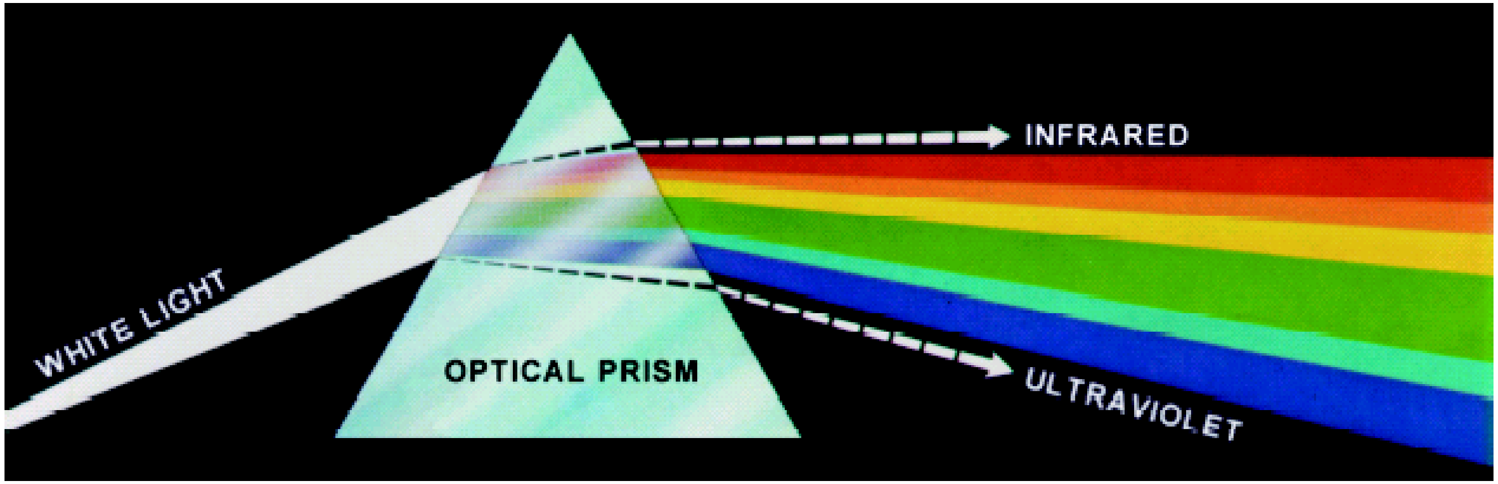
Some questions

1. What does it mean when we say an object is in a certain color?
2. Why are the primary colors of human vision red, green, and blue?
3. Is it true that different portions of red, green, and blue can produce all the visible color?
4. What kind of color model is the most suitable one to describe human vision?

Color Fundamentals

In 1666, Sir Isaac Newton discovered that:

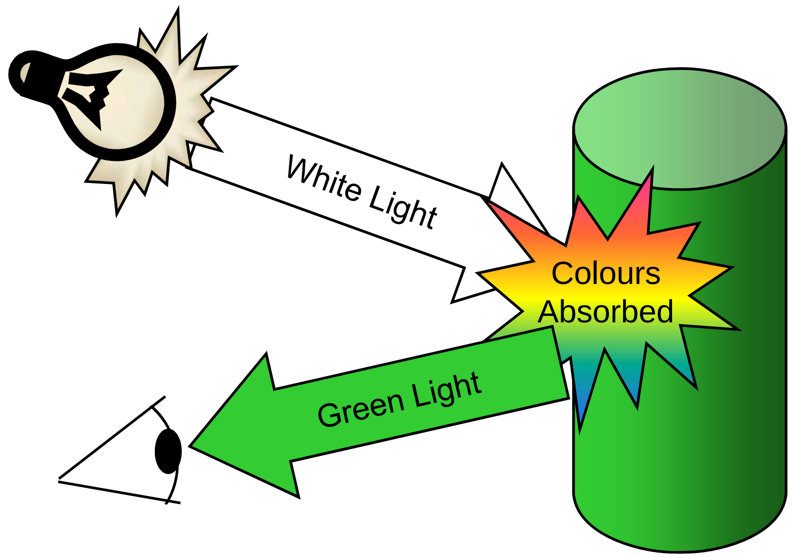
when a beam of sunlight passes through a glass prism, the emerging beam is split into a spectrum of colors



Color Fundamentals (cont…)

The colors that humans and most animals perceive in an object are determined by the nature of the light reflected from the object.

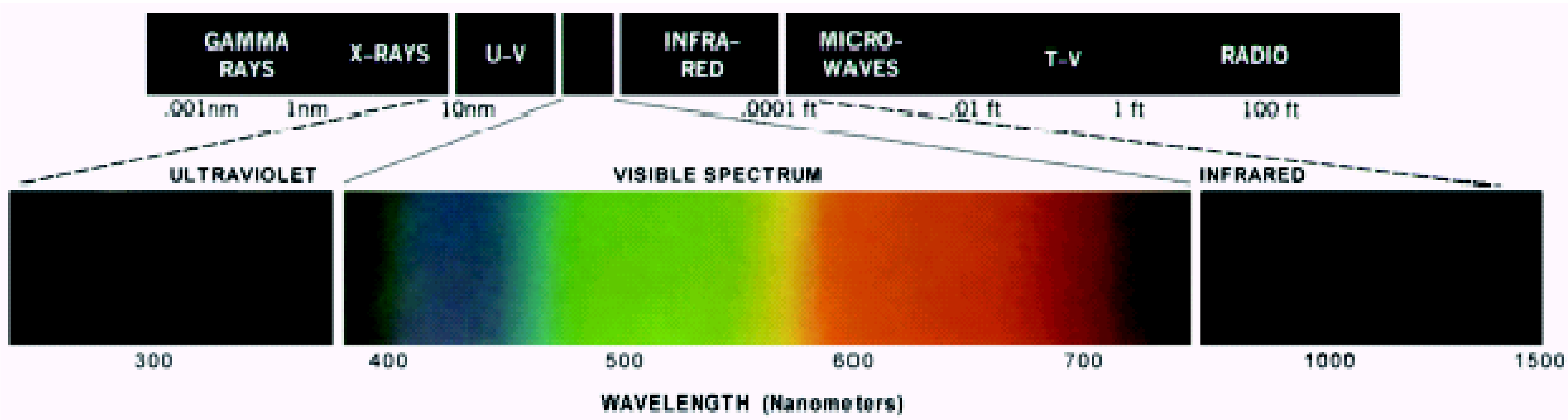
For example, green objects reflect light with wave lengths primarily in the range of 500 – 570 nm while absorbing most of the energy at other wavelengths



Color Fundamentals (cont…)

Chromatic light spans the electromagnetic spectrum from approximately 400 to 700 nm

As we mentioned before human colour vision is achieved through 6 to 7 million cones in each eye



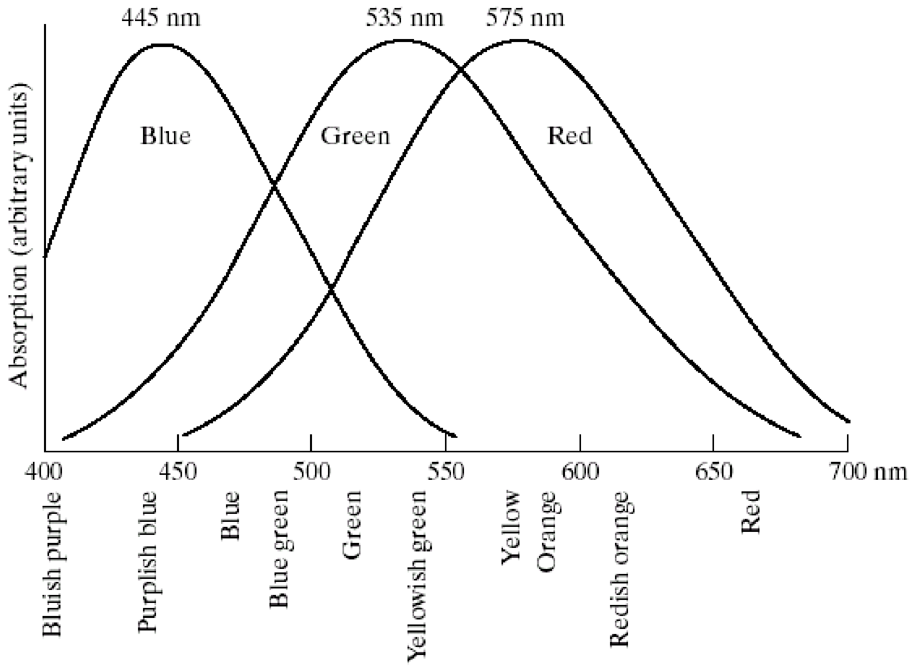
Color Fundamentals (cont…)

3 basic qualities are used to describe the quality of **a chromatic light source**:

- **Radiance:** the total amount of energy that flows from the light source (measured in watts)
- **Luminance:** the amount of energy an observer *perceives* from the light source (measured in lumens)
  - Note we can have high radiance, but low luminance
- **Brightness:** a subjective (practically unmeasurable) notion that embodies the intensity of light

Color Fundamentals (cont…)

Detailed experimental curve available in 1965



- Cones are divided into three sensible categories
  - 65% of cones are sensitive to red light
  - 33% are sensitive to green light
  - 2% are sensitive to blue light

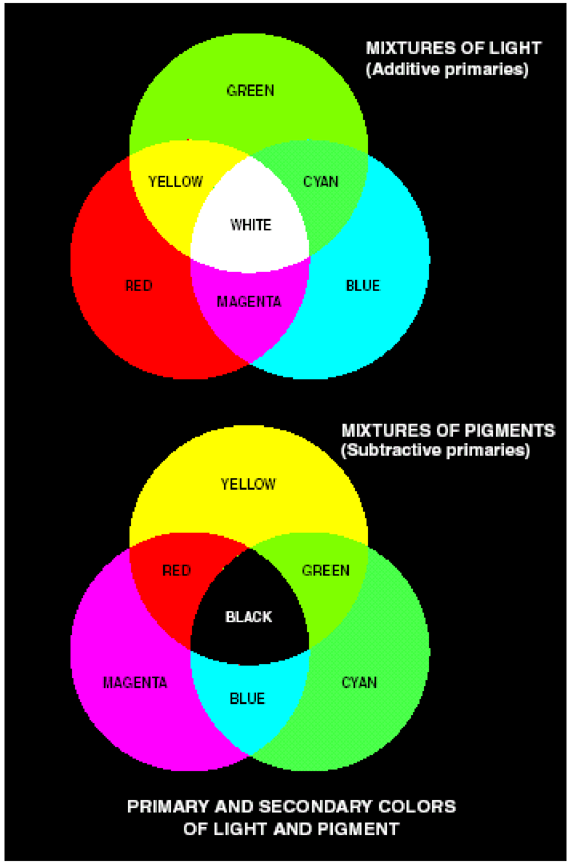
- For this reason, red, green, and blue are referred to as the primary colors of human vision. CIE standard designated three specific wavelength to these three colors in 1931.
  - Red (R) = 700 nm
  - Green (G) = 546.1 nm
  - Blue (B) = 435.8 nm

I. No single color may be called red, green, or blue.
II. R, G, B are only specified by standard.

Color Fundamentals (cont…)

Primary colors of pigment

- A primary color of pigment refers to one that absorbs the primary color of the light, but reflects the other two.
- Primary color of pigments are magenta, cyan, and yellow
- Secondary color of pigments are then red, green, and blue



a
b

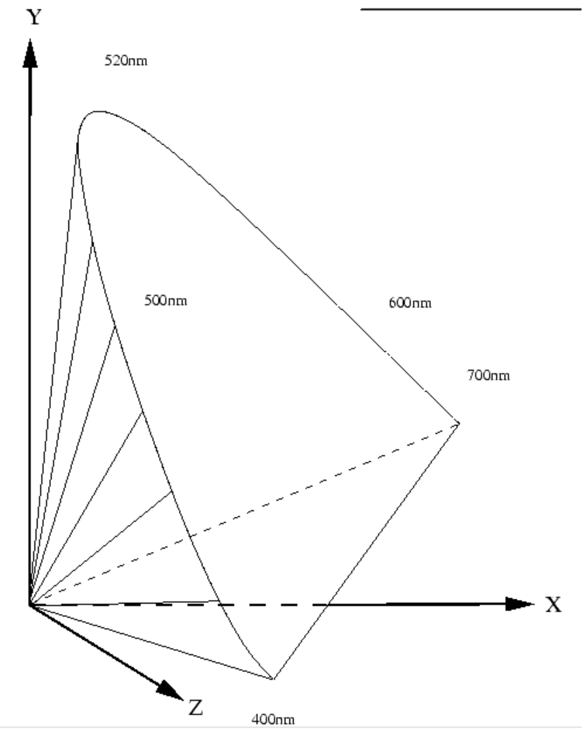
FIGURE 6.4 Primary and secondary colors of light and pigments. (Courtesy of the General Electric Co., Lamp Business Division.)

CIE Chromacity Diagram

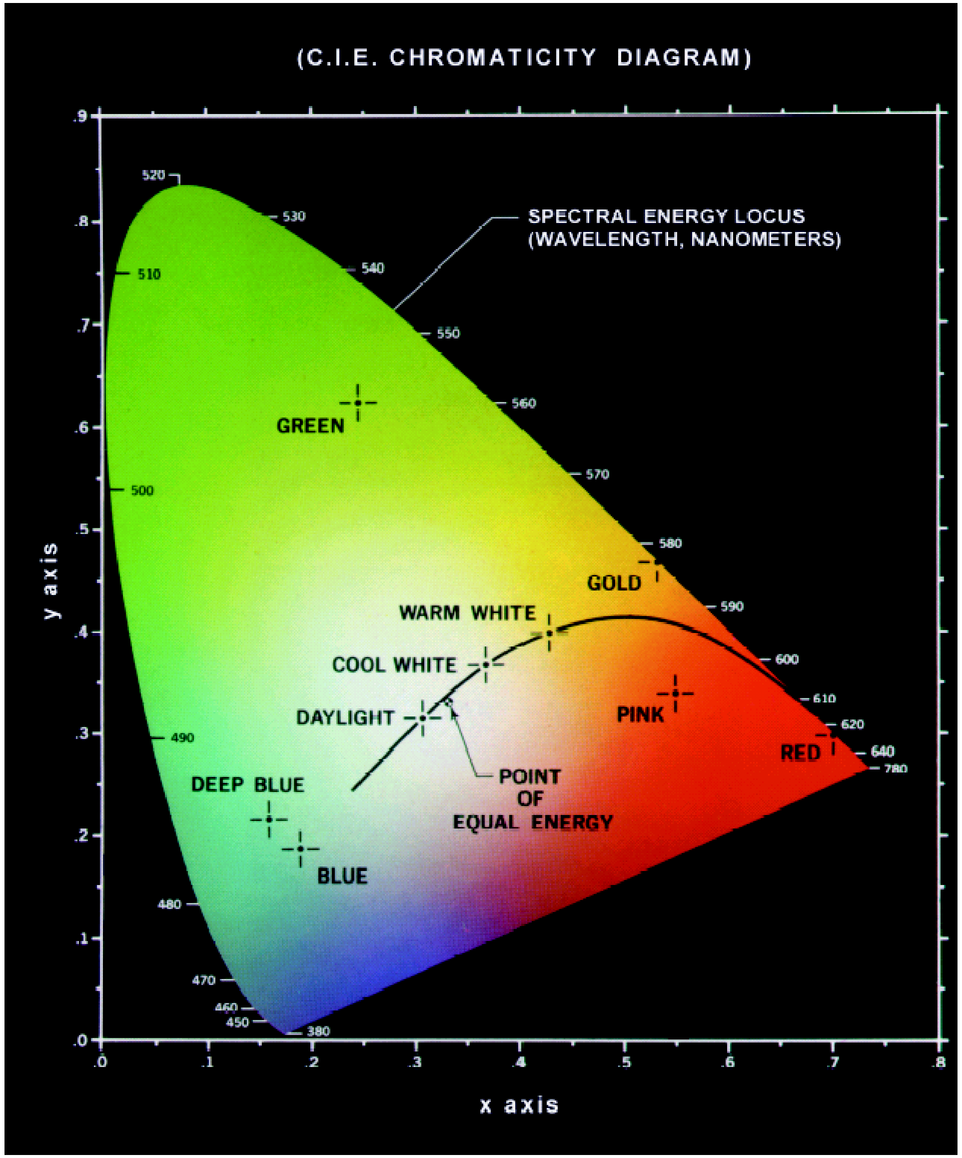
Specifying colors systematically can be achieved using **the CIE chromacity diagram.**

- x-axis represents the proportion of **red**;
- y-axis represents the proportion of **green** used.
- The proportion of **blue** used in a color is calculated as:

$$z = 1 - (x + y)$$



CIE Chromacity Diagram (cont…)



Green: 62% green, 25% red and 13% blue.

Red: 32% green, 67% red and 1% blue.

CIE Chromacity Diagram (cont…)

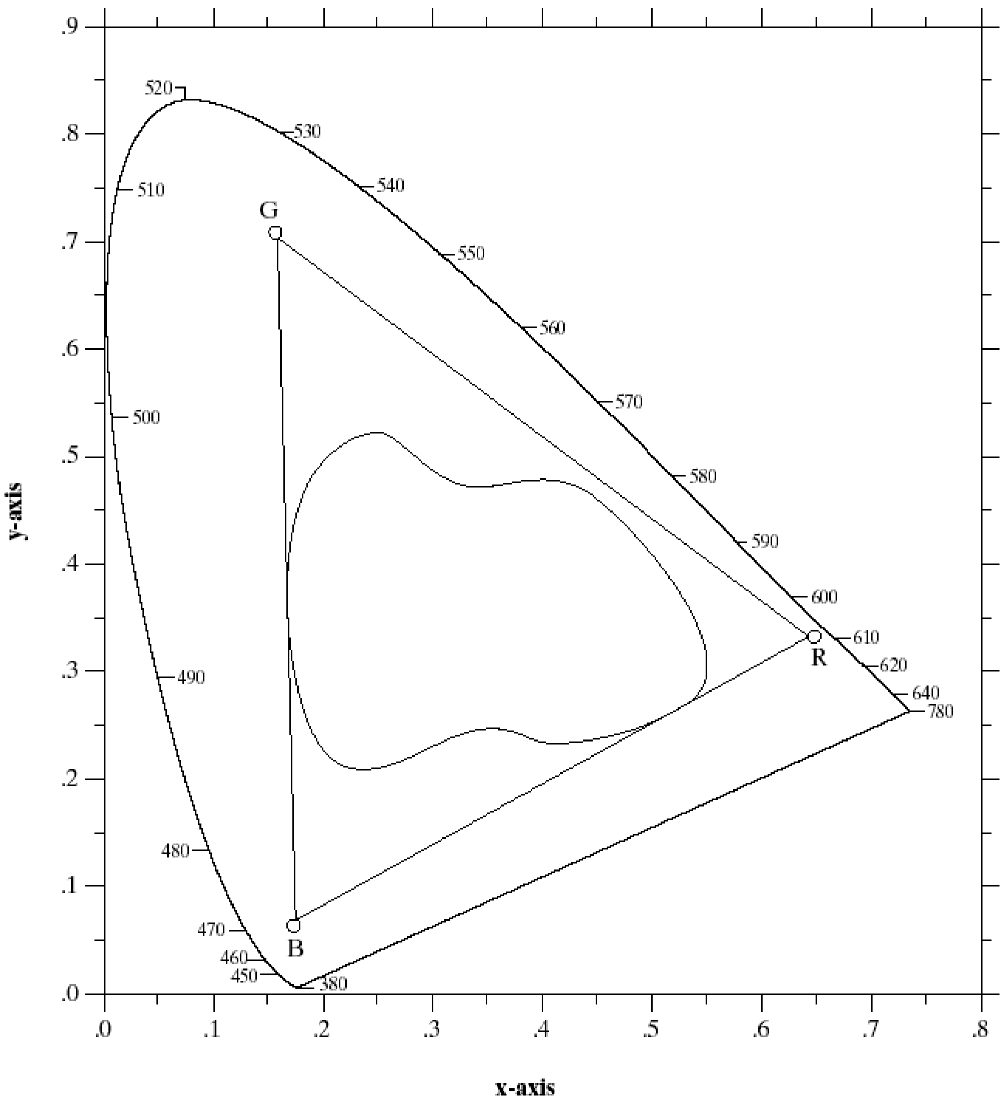
Conclusions

- Any color located on the boundary of the chromacity chart is fully saturated.

- The point of equal energy has equal amounts of each color and is the CIE standard for pure white.

- Any straight line joining two points in the diagram defines all of the different colors that can be obtained by combining these two colors additively.
- This can be easily extended to three points.

CIE Chromacity Diagram (cont…)



- This means the entire color range cannot be displayed based on any three colors.
- The triangle shows the typical color gamut produced by RGB monitors.
- The strange shape is the gamut achieved by high quality color printers.

Color Models

From the previous discussion it should be obvious that there are different ways to model color

We will consider two very popular models used in color image processing:

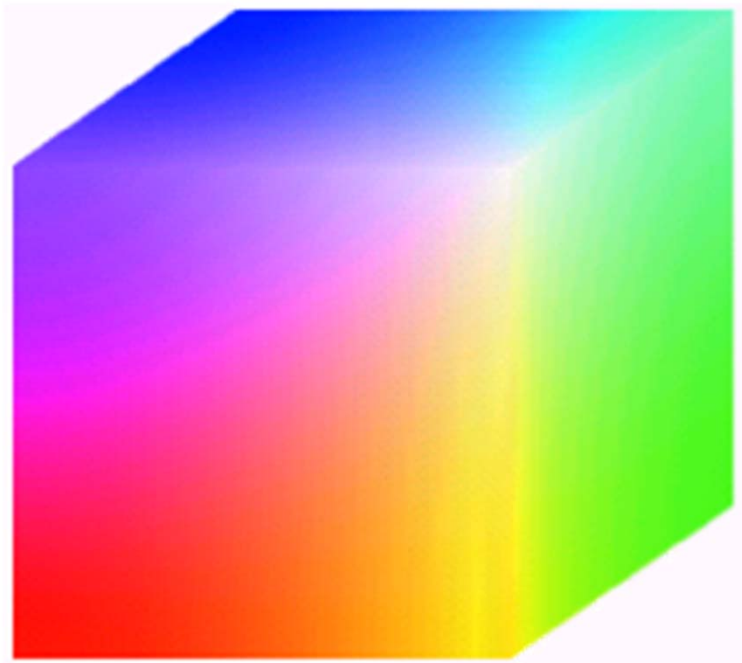
- RGB (**R**ed **G**reen **B**lue)
- HIS (**H**ue **S**aturation **I**ntensity)

RGB

Each color appears in its primary spectral components of red, green and blue.

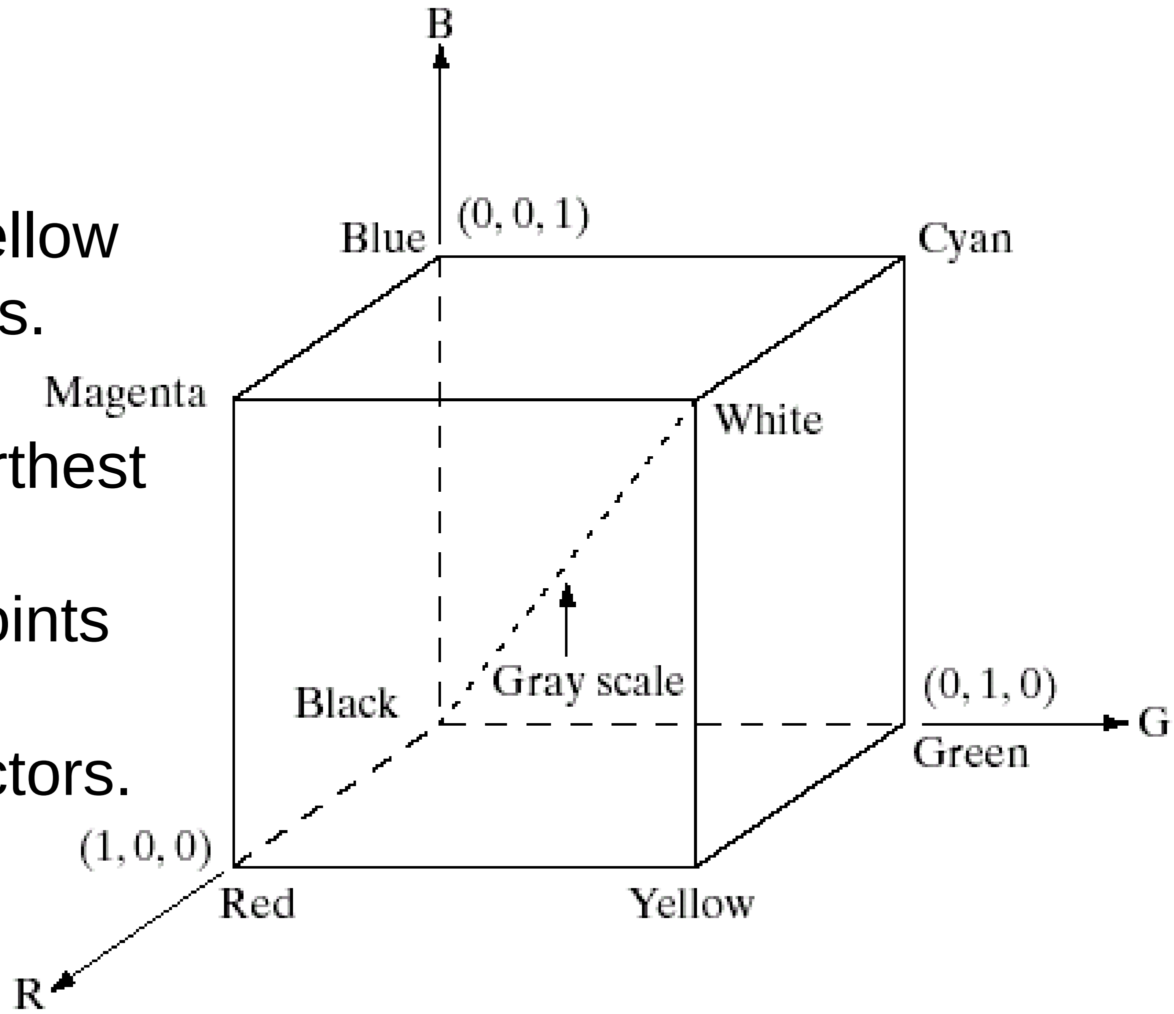
The model is based on a Cartesian coordinate system.

- Color monitor, color video cameras (additive color system)
- Pixel depth – nr of bits used to represent each pixel
  - Full color image (24 bits)



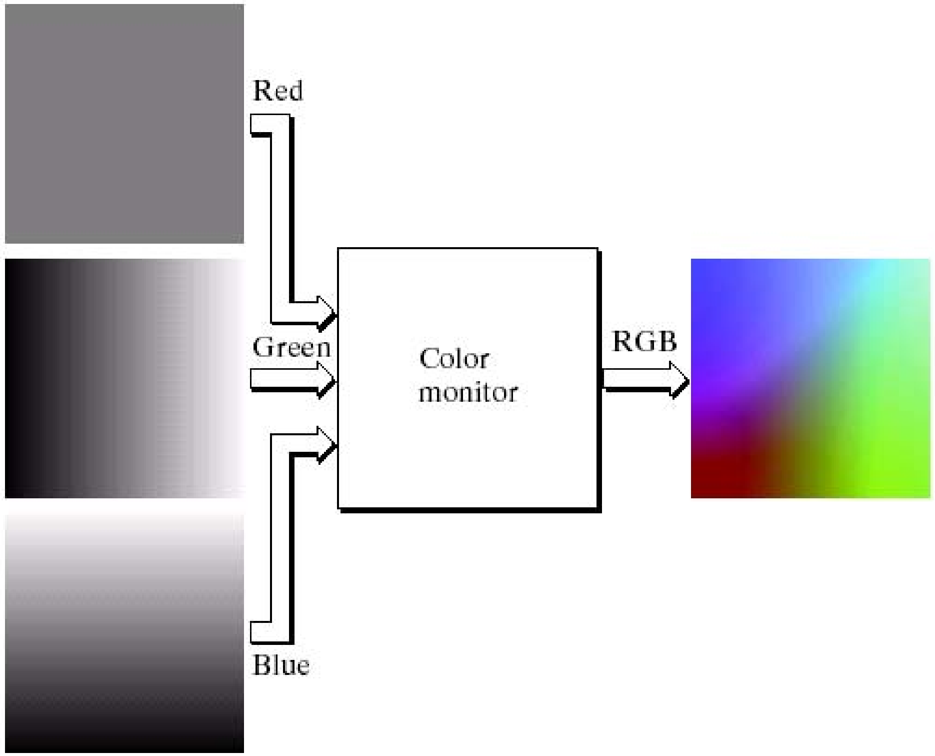
RGB (cont…)

- RGB values are at 3 corners.
- Cyan magenta and yellow are at three other corners.
- Black is at the origin.
- White is the corner furthest from the origin.
- Different colors are points on or inside the cube represented by RGB vectors.



RGB (cont…)

- When fed into a monitor RGB images are combined to create a composite color image.
- A 24-bit image is often referred to as a full-color image as it allows = 16,777,216 colors.



The HSI Color Model

- RGB is useful for hardware implementations and is serendipitously related to the way in which the human visual system works .
- However, RGB is not a particularly intuitive way in which to describe colors.
- Rather when people describe colors they tend to use **hue**, **saturation** and **brightness**.
- RGB is great for color generation, but HSI is great for color description.

The HSI Colour Model (cont…)

The HSI model uses three measures:

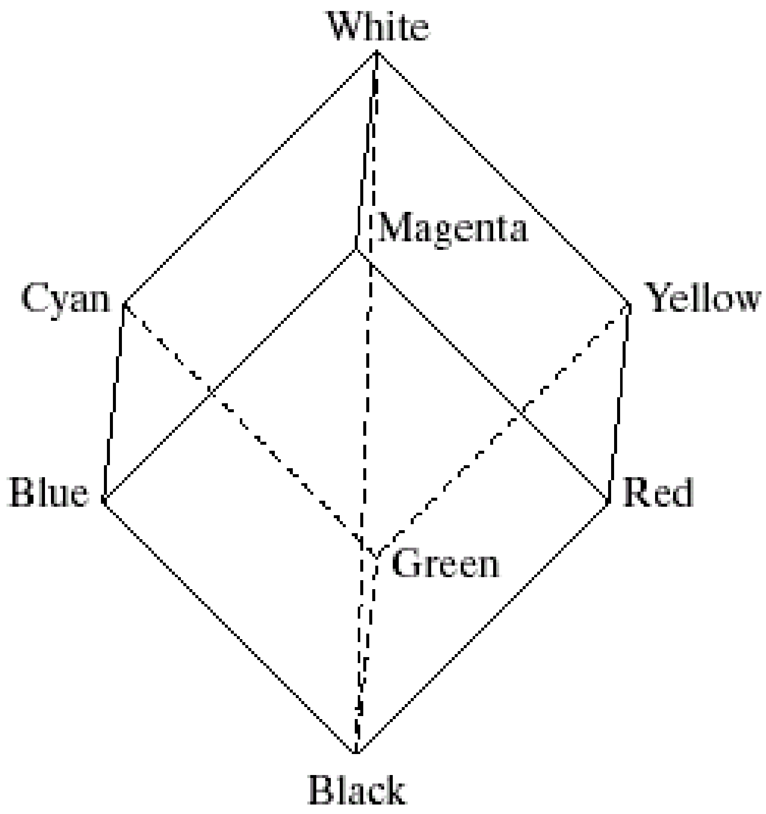
- **Hue:** A color attribute that describes a pure color (pure yellow, orange or red)
- **Saturation:** Gives a measure of how much a pure color is diluted with white light
- **Intensity:** Brightness is nearly impossible to measure because it is so subjective. Instead we use intensity. Intensity is the same achromatic notion that we have seen in grey level images

HSI, Intensity & RGB

- Intensity can be extracted from RGB images.
- Now consider if we stand RGB color cube on the black vertex and position the white vertex directly above it

The intensity component of any color can be determined by passing a plane *perpendicular* to the intensity axis and containing the color point.

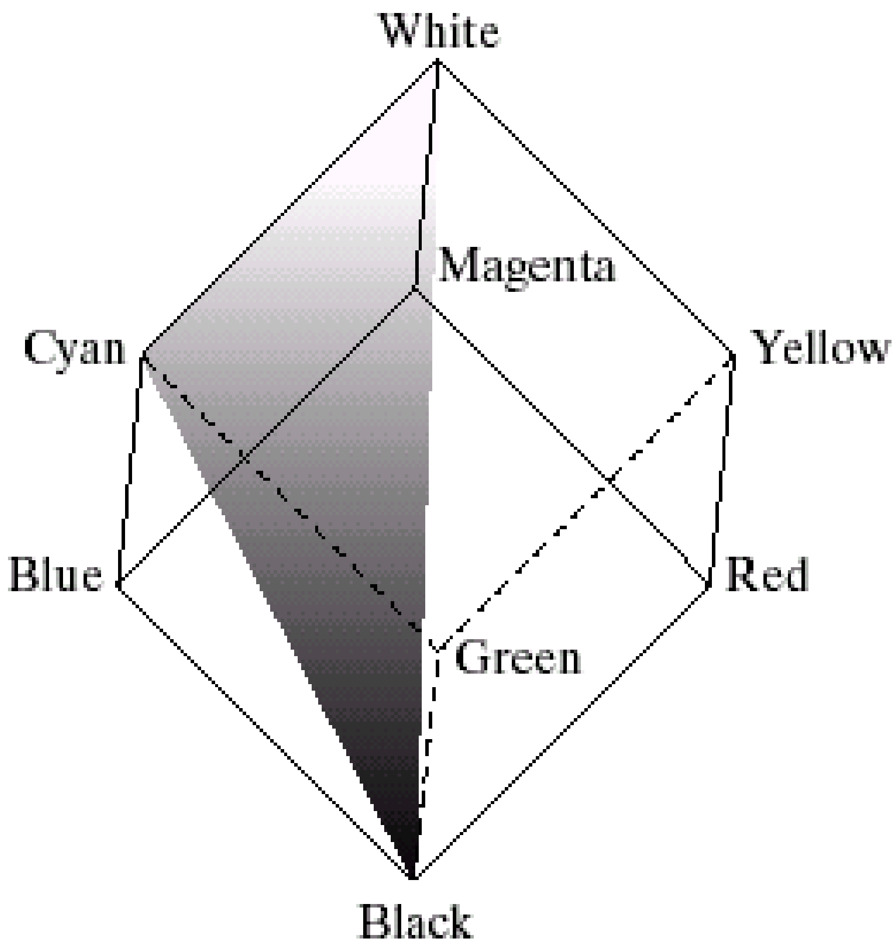
The intersection of the plane with the intensity axis gives us the intensity component of the color.



HSI, Hue & RGB

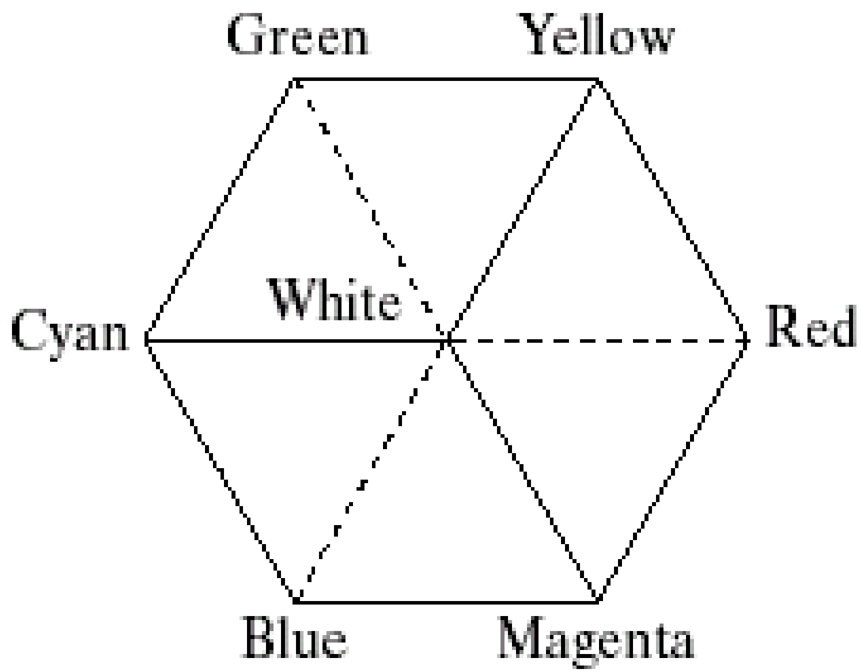
In a similar way we can extract the **hue** from the RGB color cube.

- Consider a plane defined by the three points cyan, black and white.
- All points contained in this plane must have the same hue (cyan) as black and white cannot contribute hue information to a color.



The HSI Colour Model

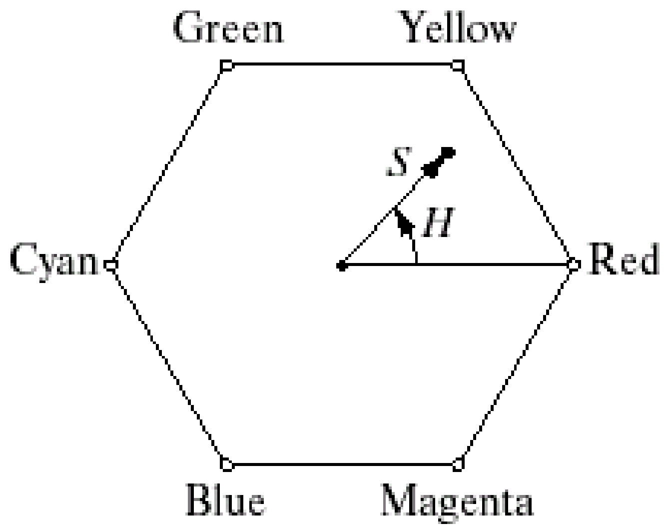
- Consider if we look straight down at the RGB cube as it was arranged previously.
- We would see a hexagonal shape with each primary color separated by 120° and secondary colors at 60° from the primaries.
- So the **HSI model is composed of a vertical intensity axis and the locus of color points that lie on planes perpendicular to that axis**.



The HSI Colour Model (cont…)

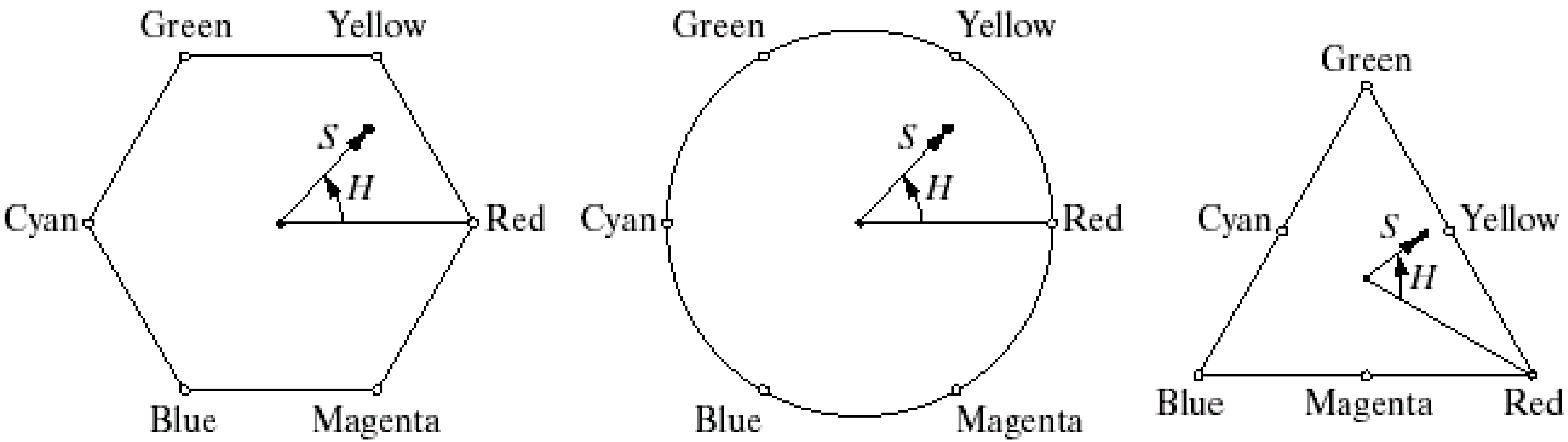
We see a hexagonal shape and an arbitrary color point

– The **hue** is determined by an angle from a reference point, usually red
– The **saturation** is the distance from the origin to the point
– The **intensity** is determined by how far up the vertical intensity axis this hexagonal plane sits (not apparent from this diagram

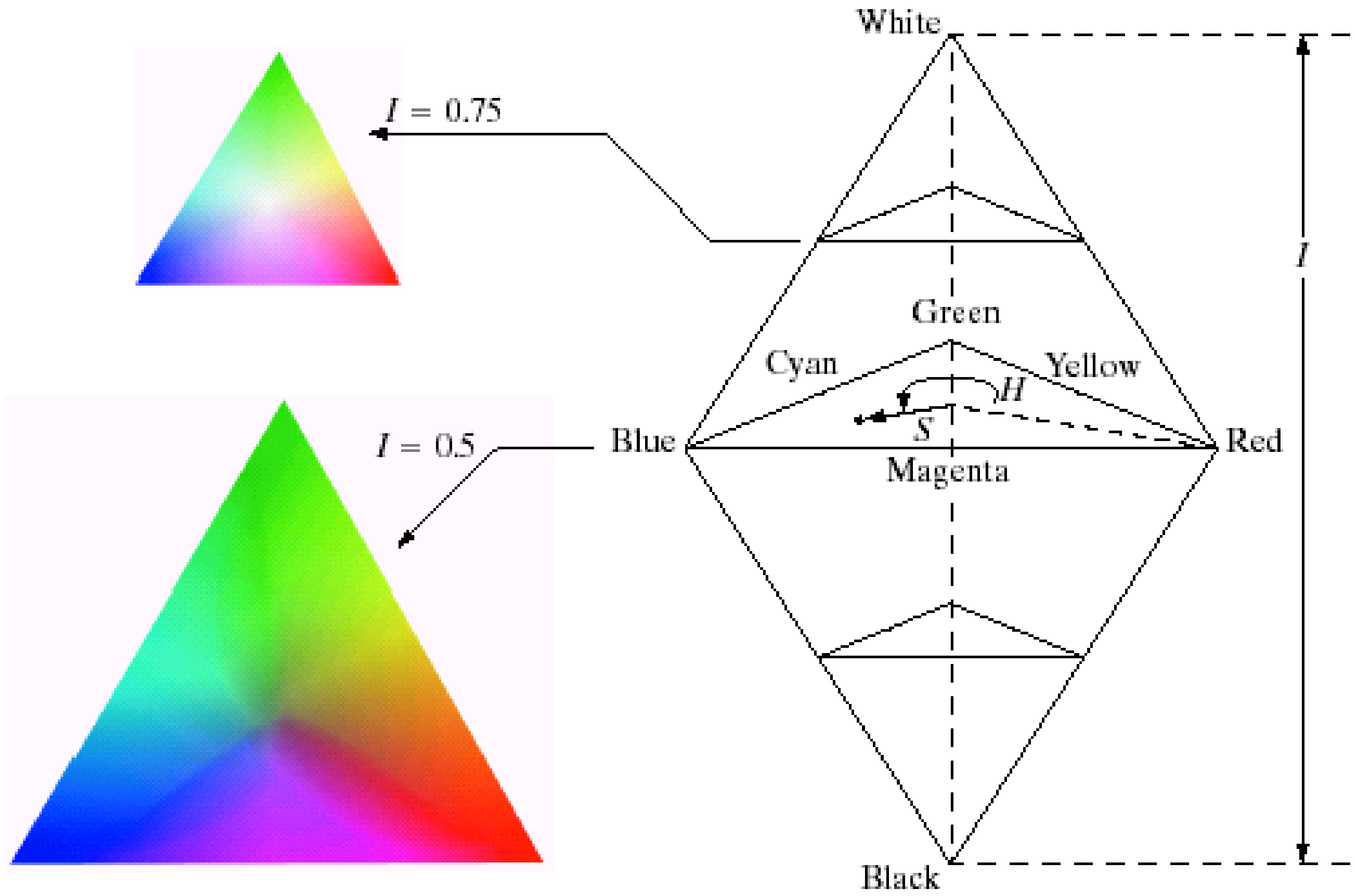


The HSI Colour Model (cont…)

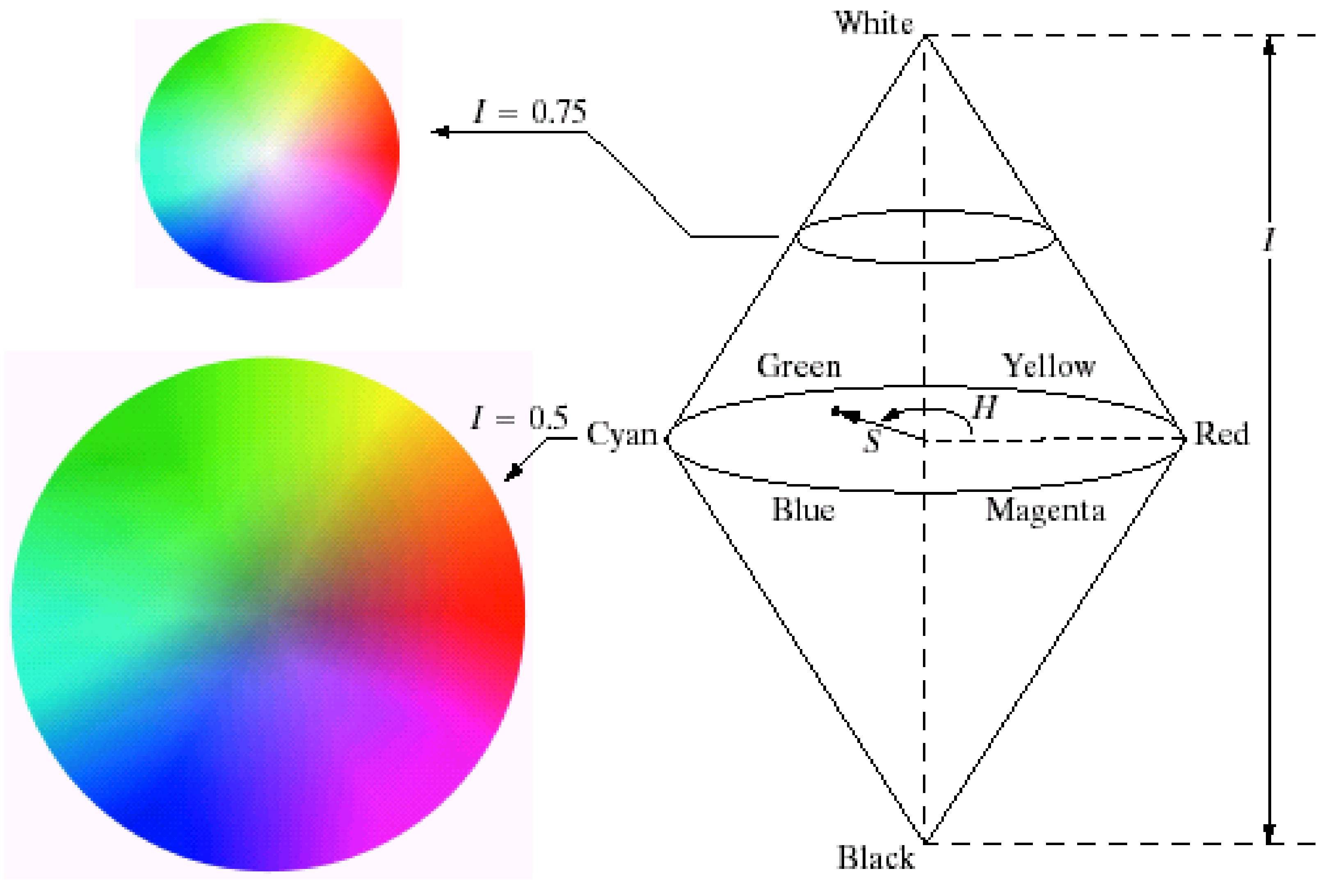
The angle and the length of the saturation vector, this plane is often represented as a circle or a triangle.



HSI Model Examples



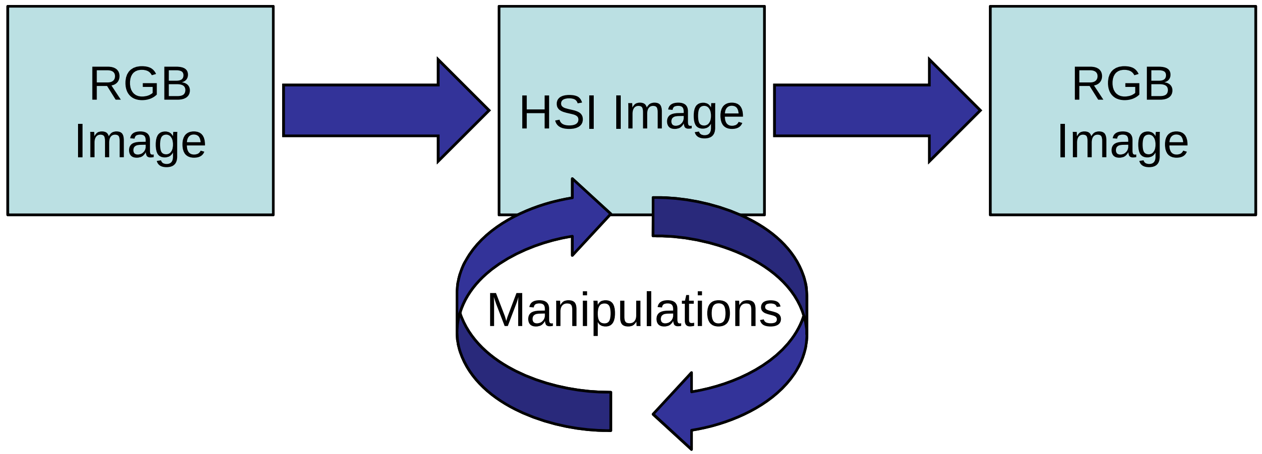
HSI Model Examples



Manipulating Images In The HSI Model

In order to manipulate an image under the HIS model we:

- First convert it from RGB to HIS
- Perform manipulations under HSI
- Finally convert the image back from HSI to RGB



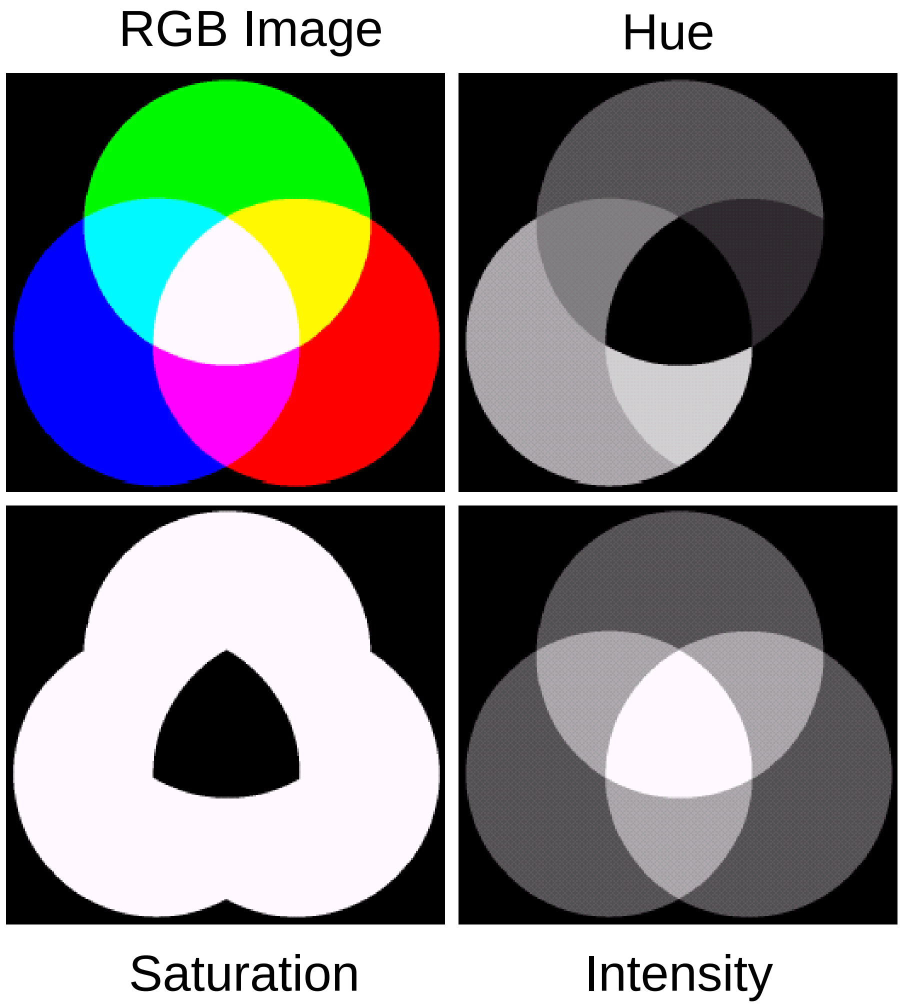
RGB -> HSI -> RGB

$$I = \frac{1}{3}(R + G + B)$$

$$S = 1 - \frac{3}{I}\min(R, G, B)$$

$$\theta = \cos^{-1}\left[\frac{\frac{1}{2}[(R-G)+(R-B)]}{\sqrt{(R-G)^2+(R-B)(G-B)}}\right]$$

$$H = \begin{cases} \theta & G \geq B \\ 2\pi - \theta & G \leq B \end{cases}$$



RGB -> HSI -> RGB (cont…)

- For 0º <= H < 120º

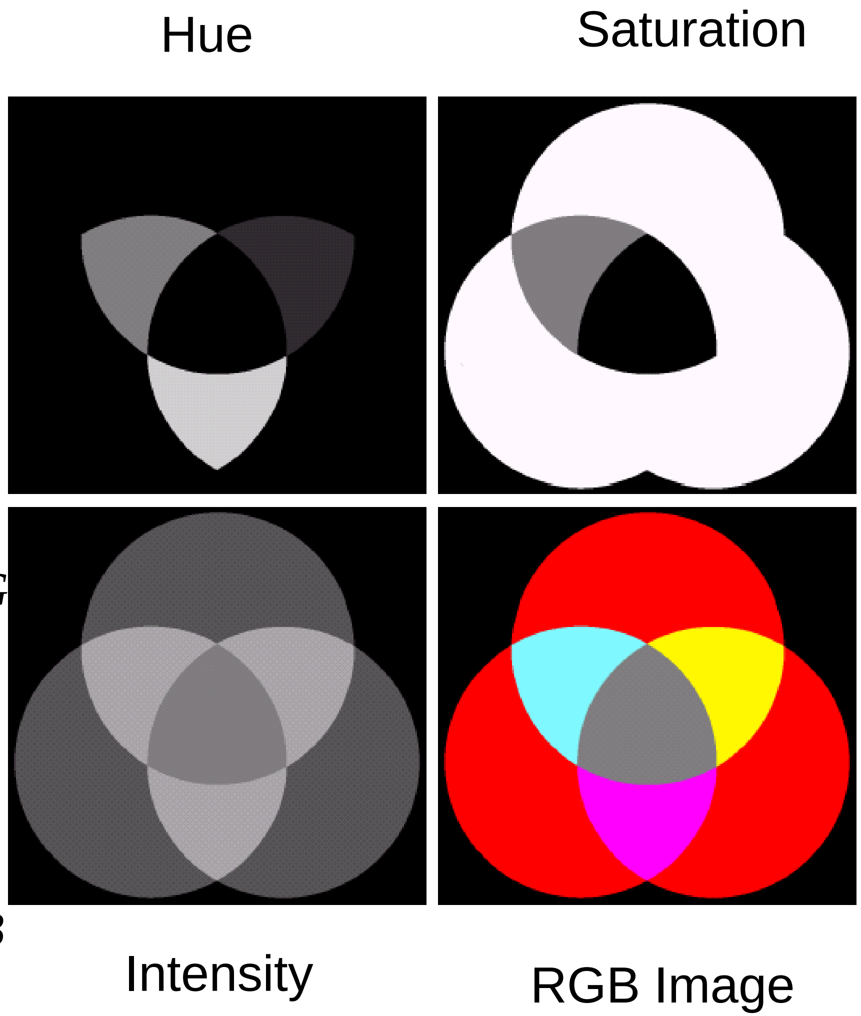
$$R = I\left[1 + \frac{S\cos(H)}{\cos(60^\circ - H)}\right], B = I(1 - S), G = I - R - B$$

- For 120º <= H < 240º

$$G = I\left[1 + \frac{S\cos(H - 120^\circ)}{\cos(180^\circ - H)}\right], R = I(1 - S), B = I - R - G$$

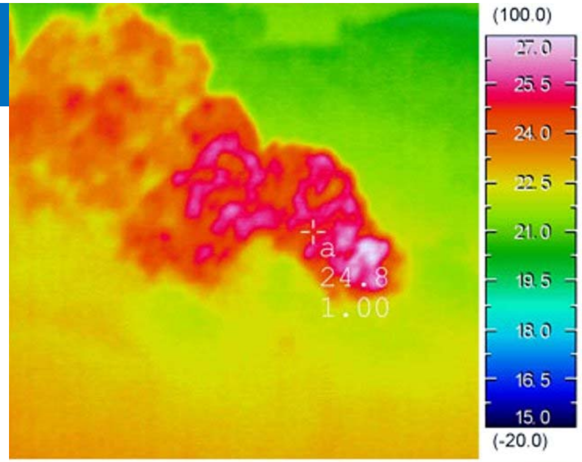
- For 240º <= H < 360º

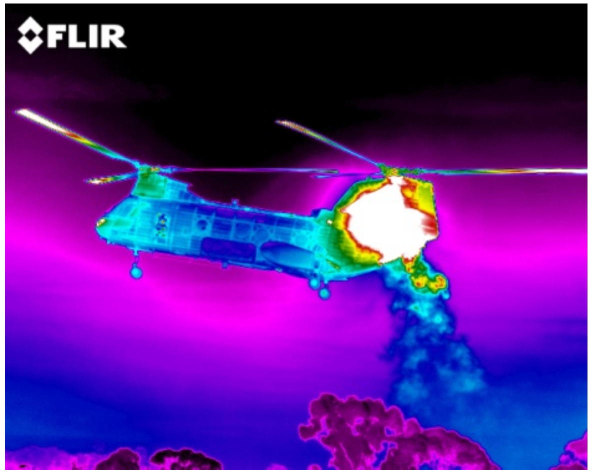
$$B = I\left[1 + \frac{S\cos(H - 240^\circ)}{\cos(300^\circ - H)}\right], G = I(1 - S), R = I - G - B$$

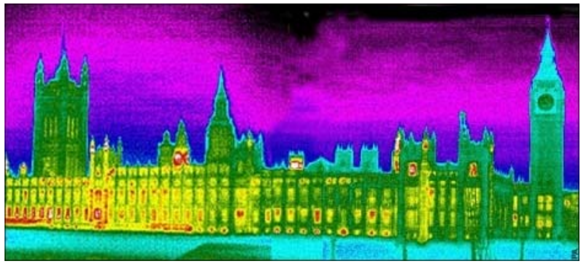


Pseudocolor Image Processing

- Pseudocolor (also called false color) image processing consists of assigning colors to grey values based on a specific criterion.
- The principle use of pseudocolor image processing is for human visualization.
  - Humans can discern between thousands of color shades and intensities, compared to only about two dozen or so shades of grey.



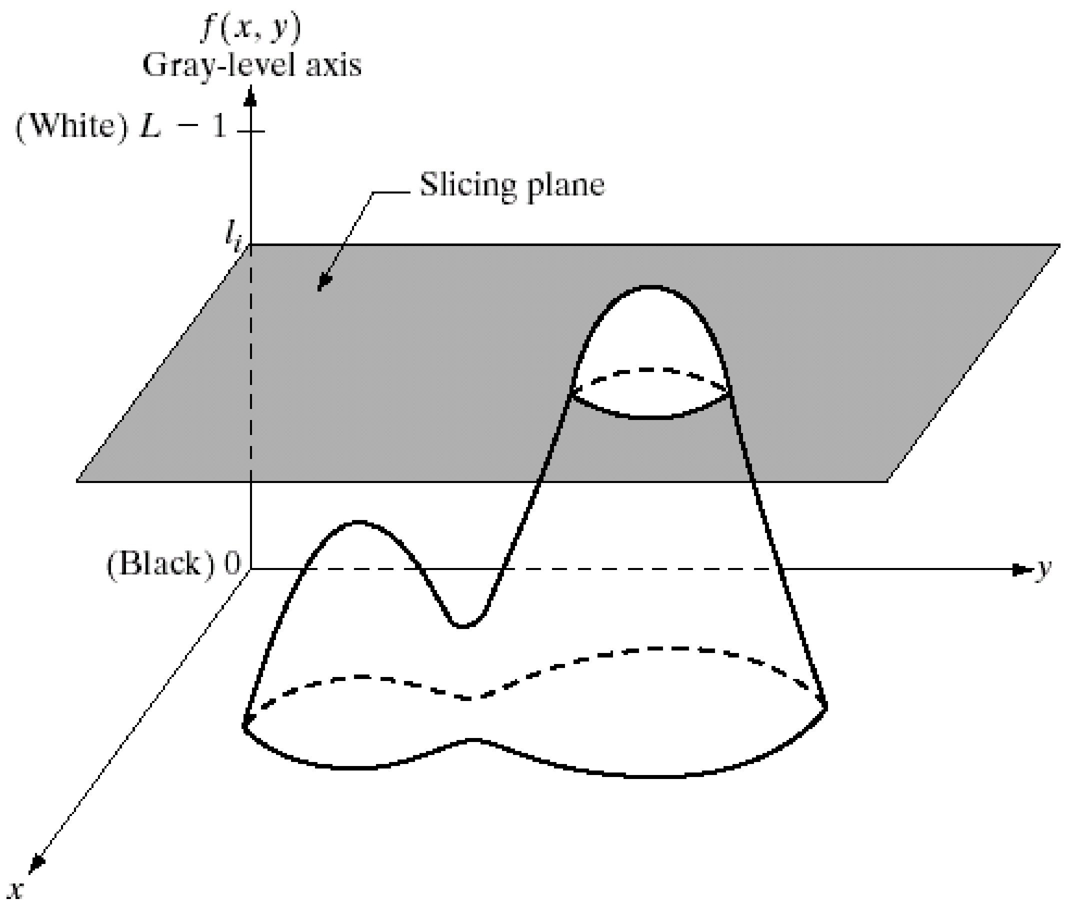


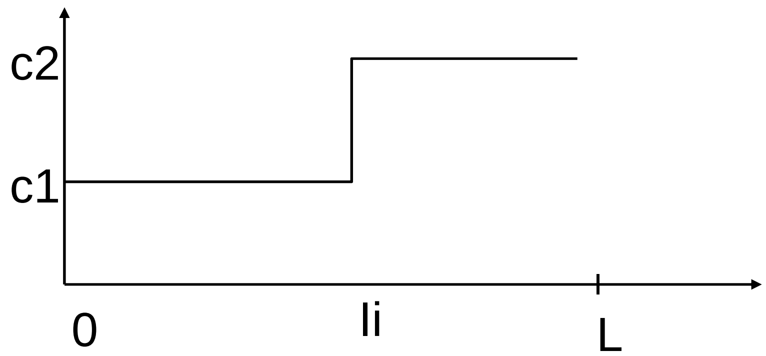


Pseudo Color Image Processing – Intensity Slicing

- Intensity slicing and color coding is one of the simplest kinds of pseudocolor image processing.
- First we consider an image as a 3D function mapping spatial coordinates to intensities .
- Now consider placing planes at certain levels parallel to the coordinate plane.
- If a value is one side of such a plane it is rendered in one color, and a different color if on the other side.

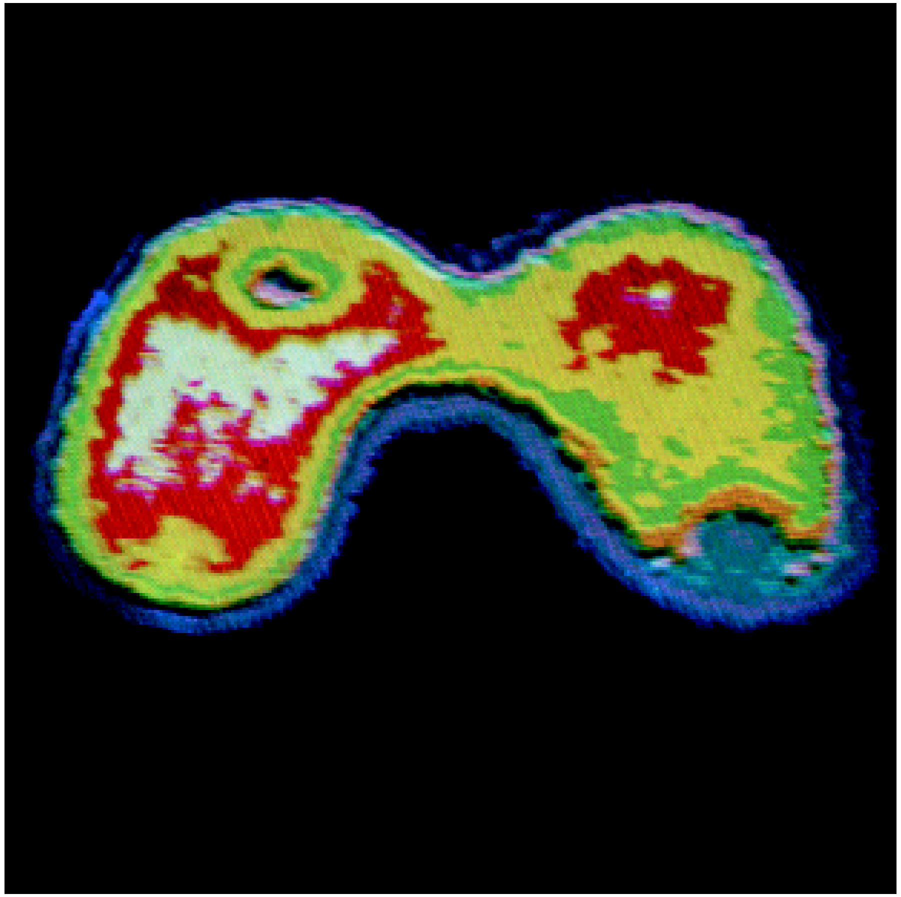
Pseudocolor Image Processing – Intensity Slicing (cont…)



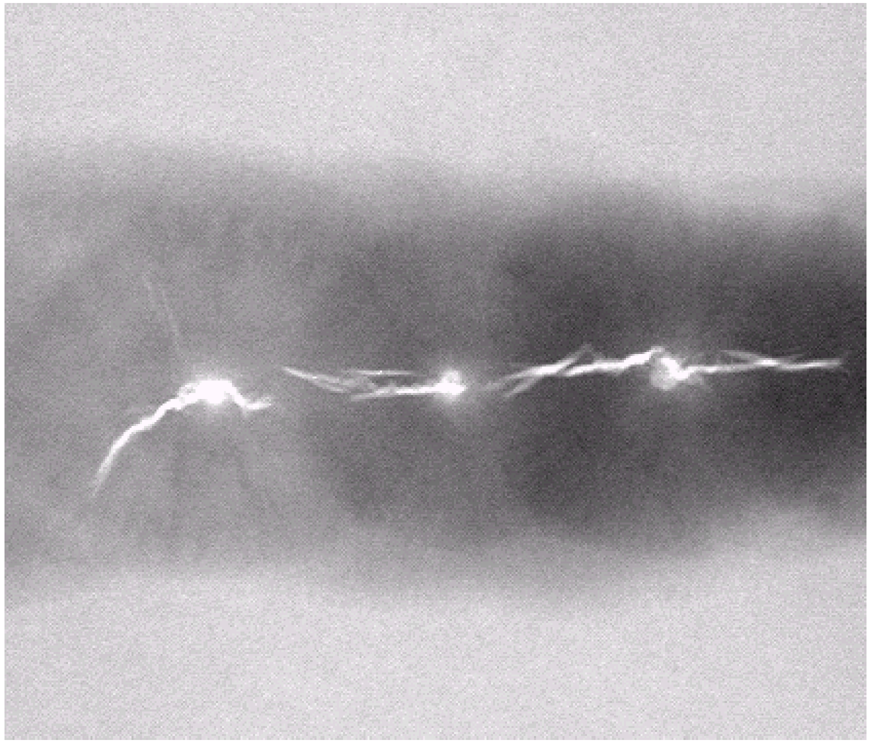


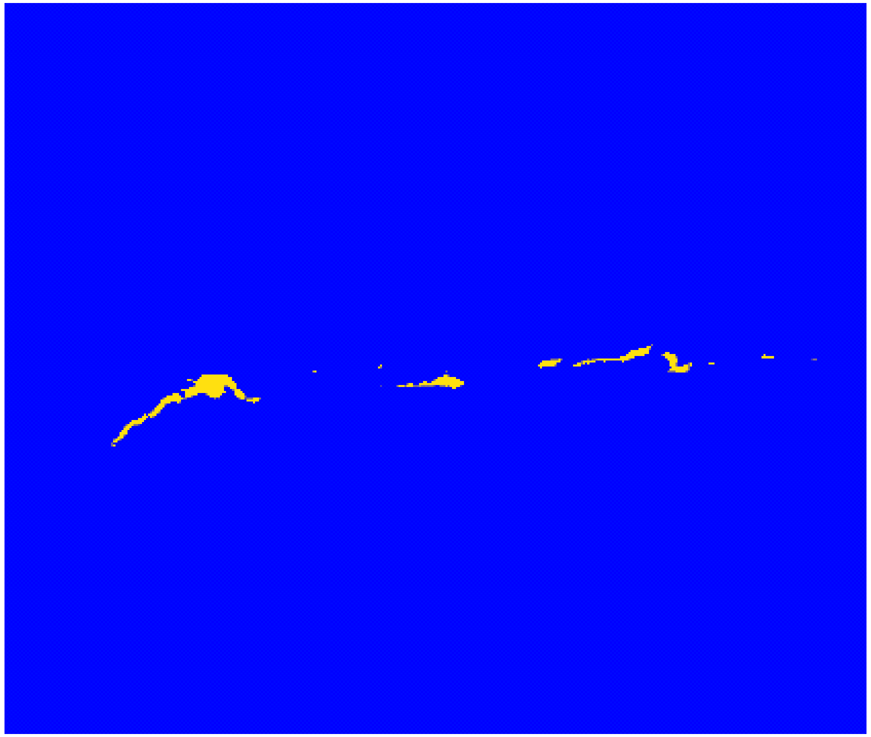
- Similar to thresholding

Pseudocolor Image Processing – Intensity Slicing (cont…)

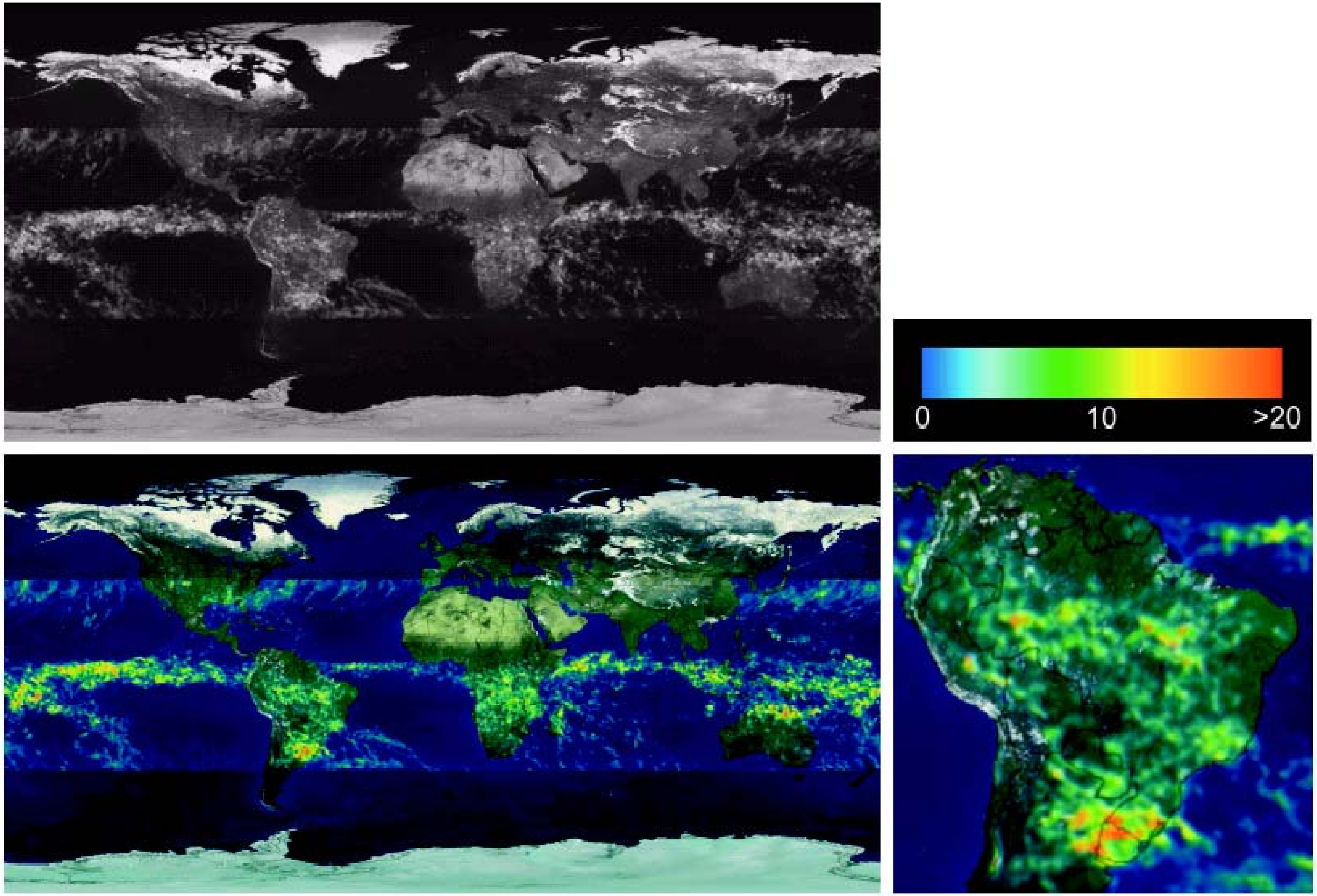


Pseudocolor Image Processing – Intensity Slicing (cont…)





Pseudocolor Image Processing – Intensity Slicing (cont…)



Pseudocolour Image Processing – Intensity to color transformations

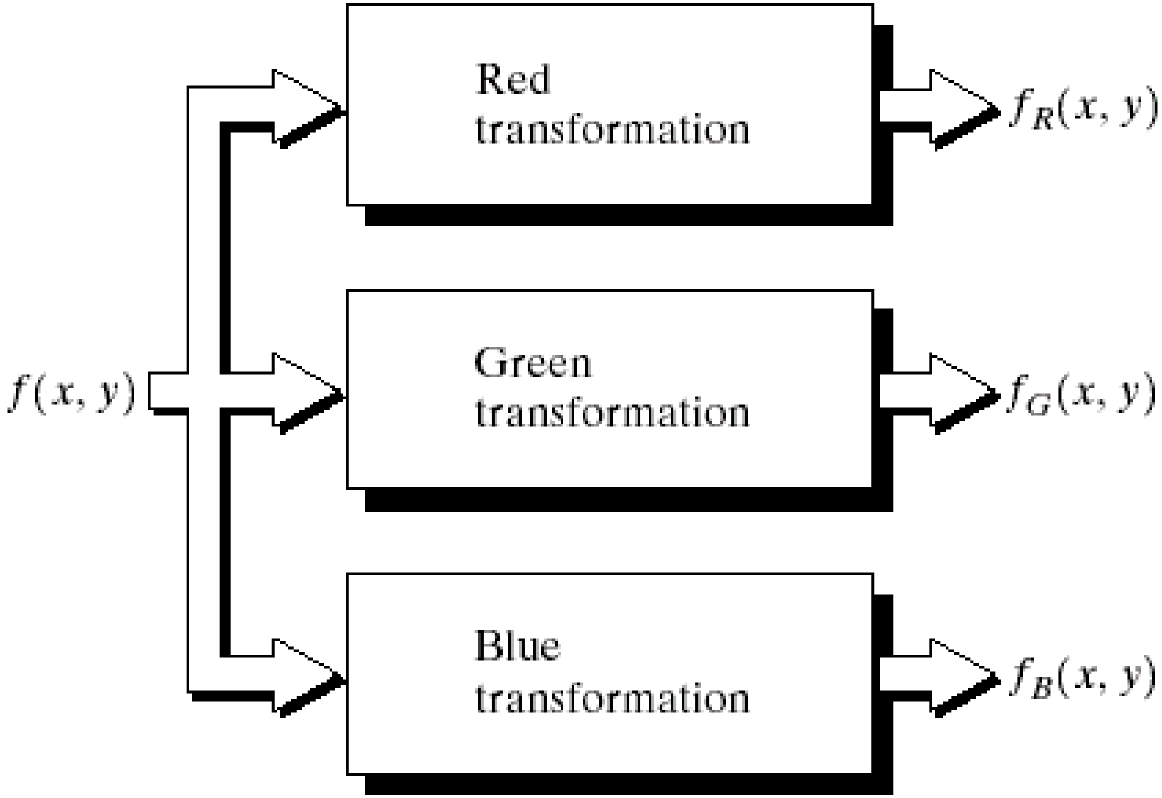


FIGURE 6.23 Functional block diagram for pseudocolor image processing. f_R, f_G, and f_B are fed into the corresponding red, green, and blue inputs of an RGB color monitor.

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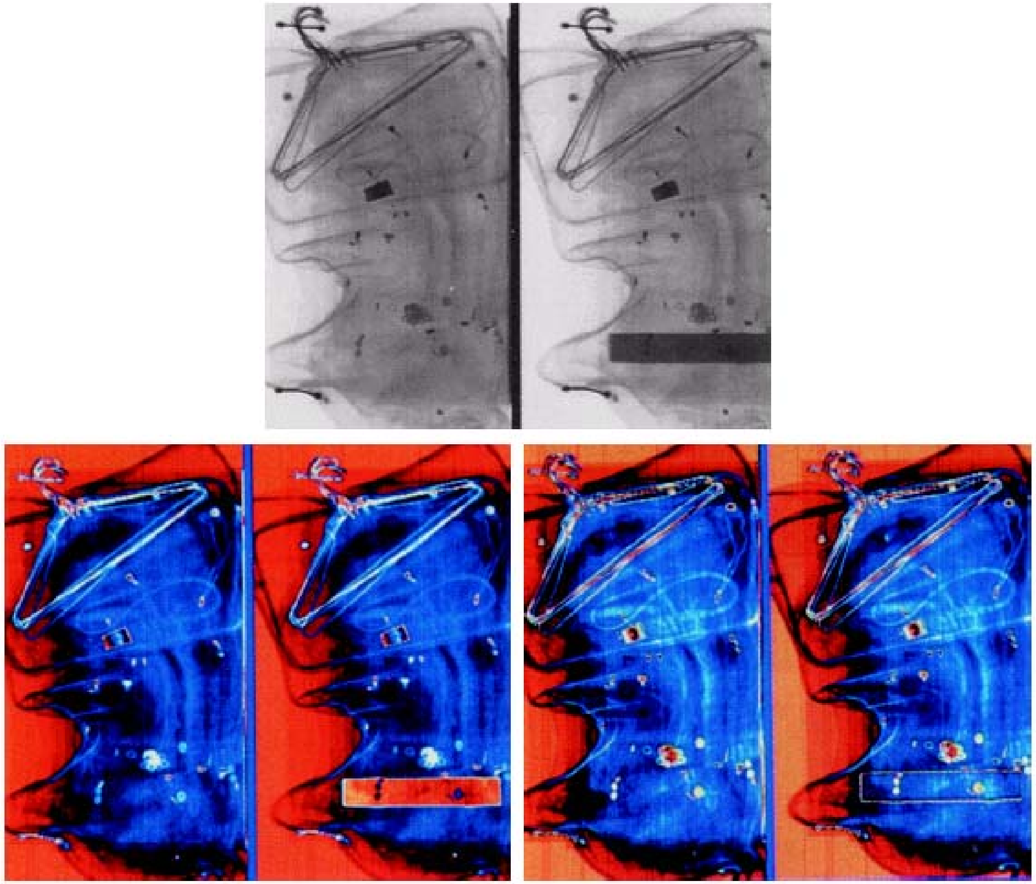
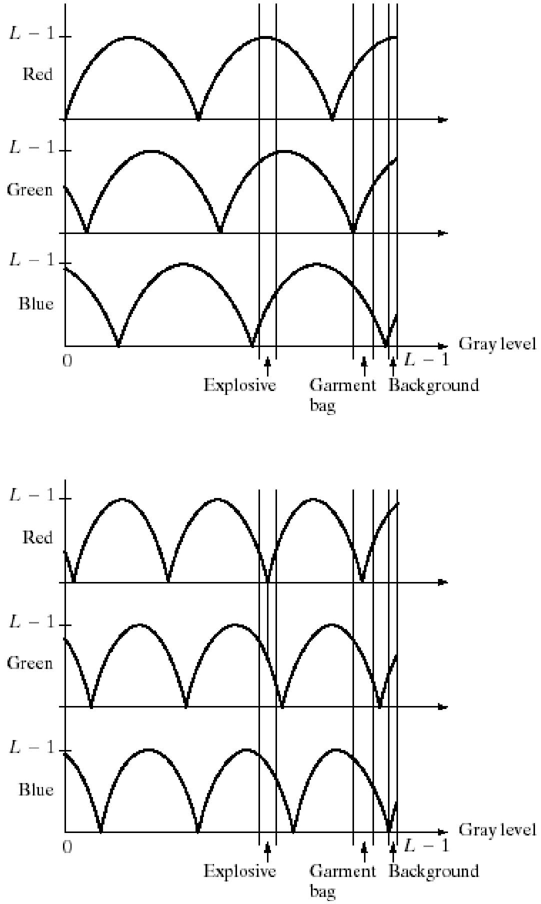
Xiamen University

Pseudocolour Image Processing – Intensity to color transformations



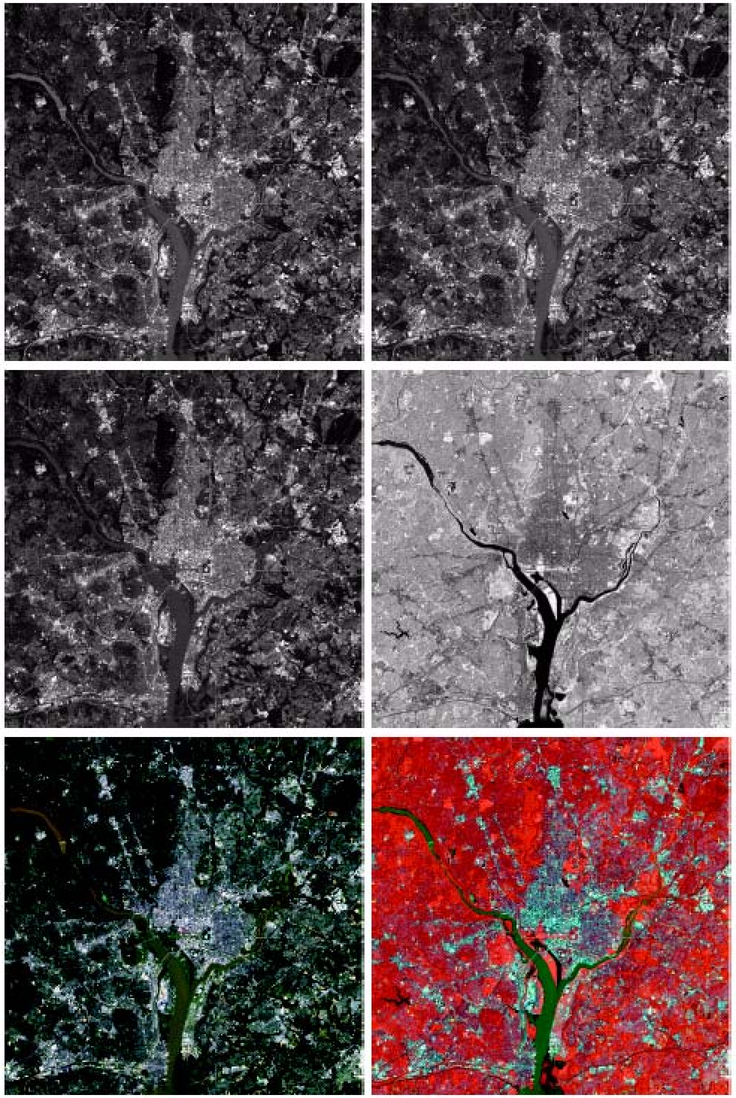
a
b c

FIGURE 6.24 Pseudocolor enhancement by using the gray-level to color transformations in Fig. 6.25. (Original image courtesy of Dr. Mike Hurwitz, Westinghouse.)



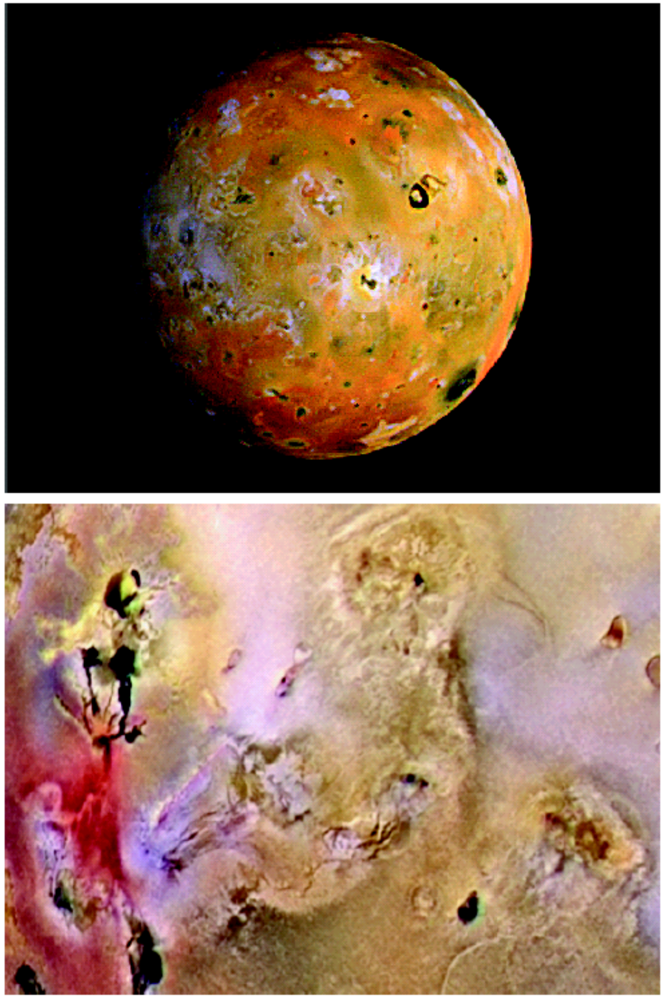
a
b

FIGURE 6.25 Transformation functions used to obtain the images in Fig. 6.24.



a b
c d
e f

FIGURE 6.27 (a)–(d) Images in bands 1–4 in Fig. 1.10 (see Table 1.1). (e) Color composite image obtained by treating (a), (b), and (c) as the red, green, blue components of an RGB image. (f) Image obtained in the same manner, but using in the red channel the near-infrared image in (d). (Original multispectral images courtesy of NASA.)



a
b

FIGURE 6.28 (a) Pseudocolor rendition of Jupiter Moon Io. (b) A close-up. (Courtesy of NASA.)

Full color processing-Color transformations

FIGURE 6.30 A full-color image and its various color-space components. (Original image courtesy of Med-Data Interactive.)

Thank you!

Full color processing-
Color transformations-color complements

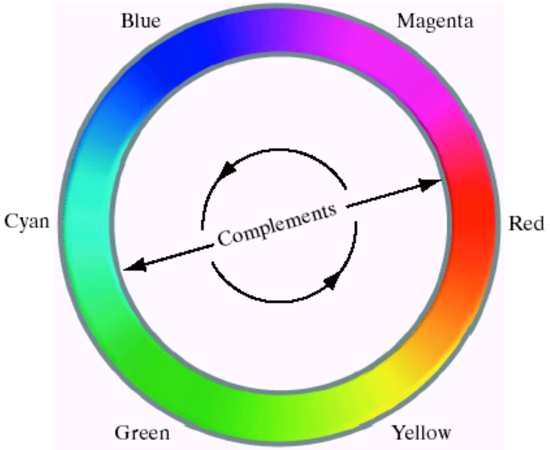
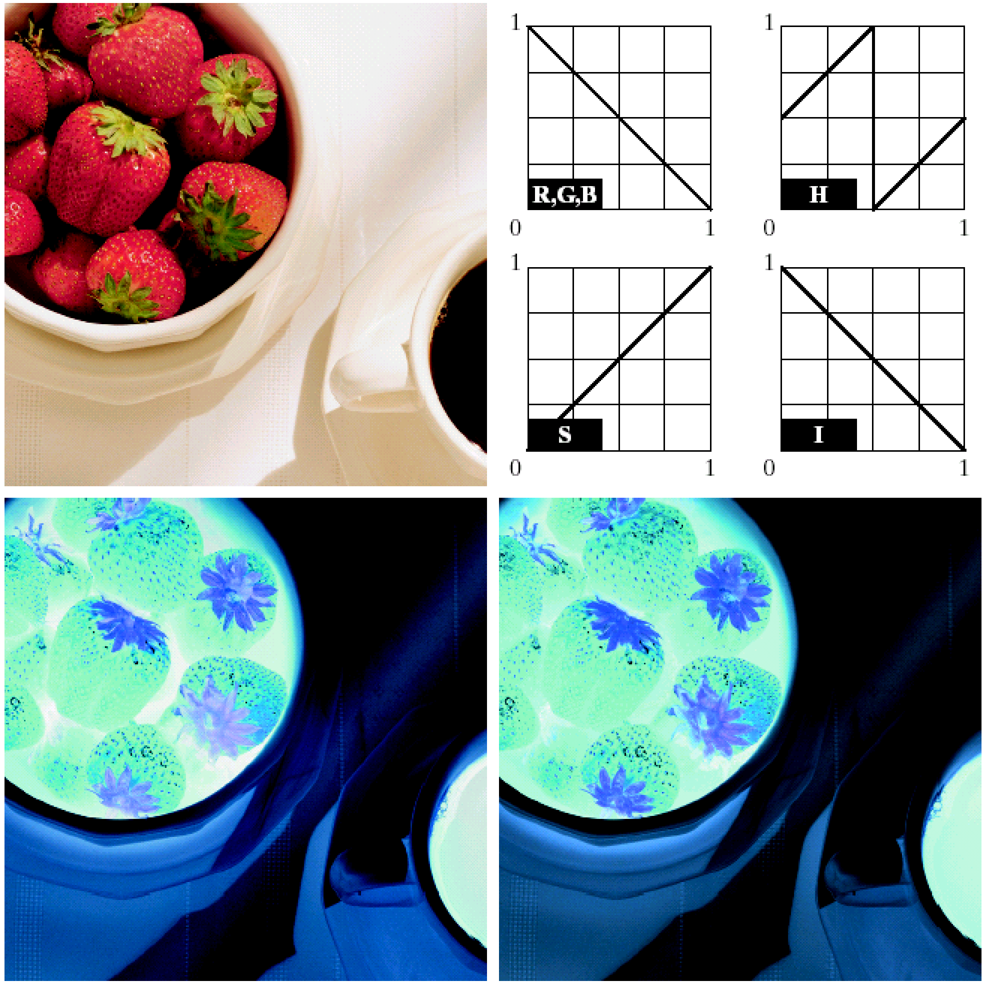


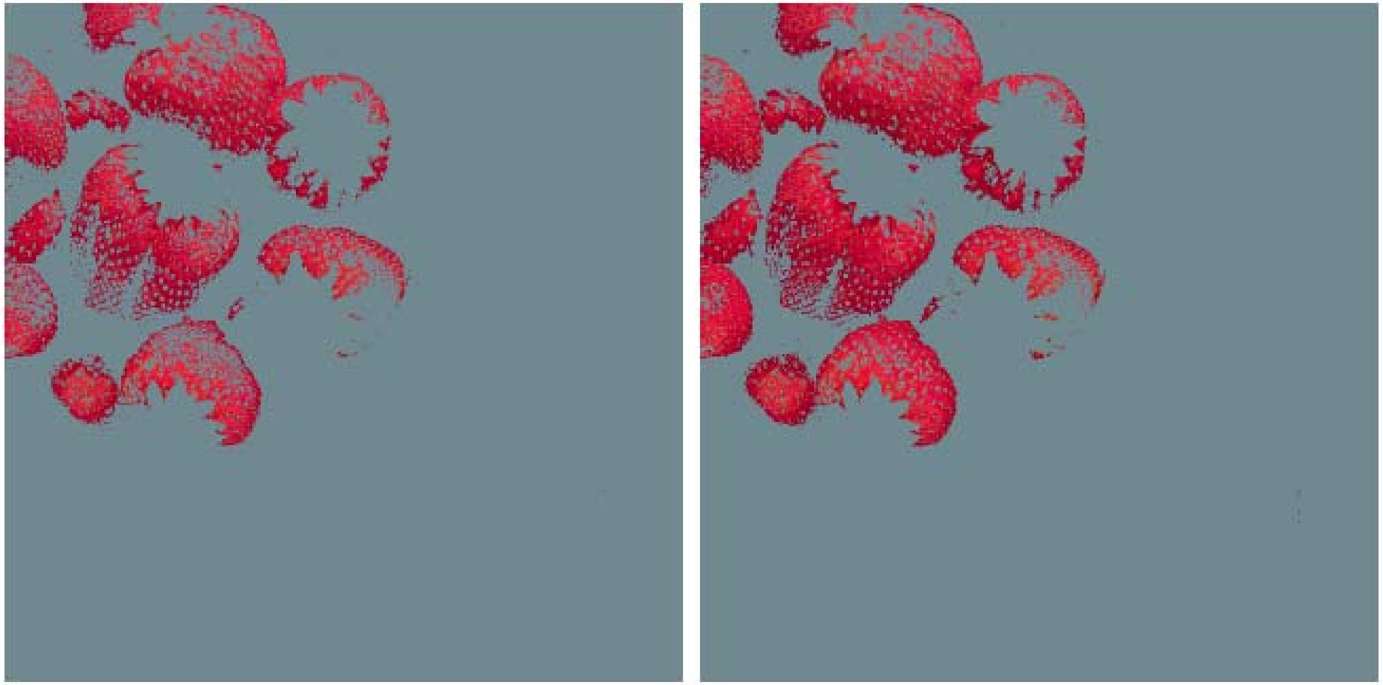
FIGURE 6.32 Complements on the color circle.



a b
c d

FIGURE 6.33 Color complement transformations. (a) Original image. (b) Complement transformation functions. (c) Complement of (a) based on the RGB mapping functions. (d) An approximation of the RGB complement using HSI transformations.

Full color processing-
Color transformations-color slicing



a b

FIGURE 6.34 Color slicing transformations that detect (a) reds within an RGB cube of width $W = 0.2549$ centered at $(0.6863, 0.1608, 0.1922)$, and (b) reds within an RGB sphere of radius 0.1765 centered at the same point. Pixels outside the cube and sphere were replaced by color $(0.5, 0.5, 0.5)$.

Full color processing-
Color transformations-tone and color corrections

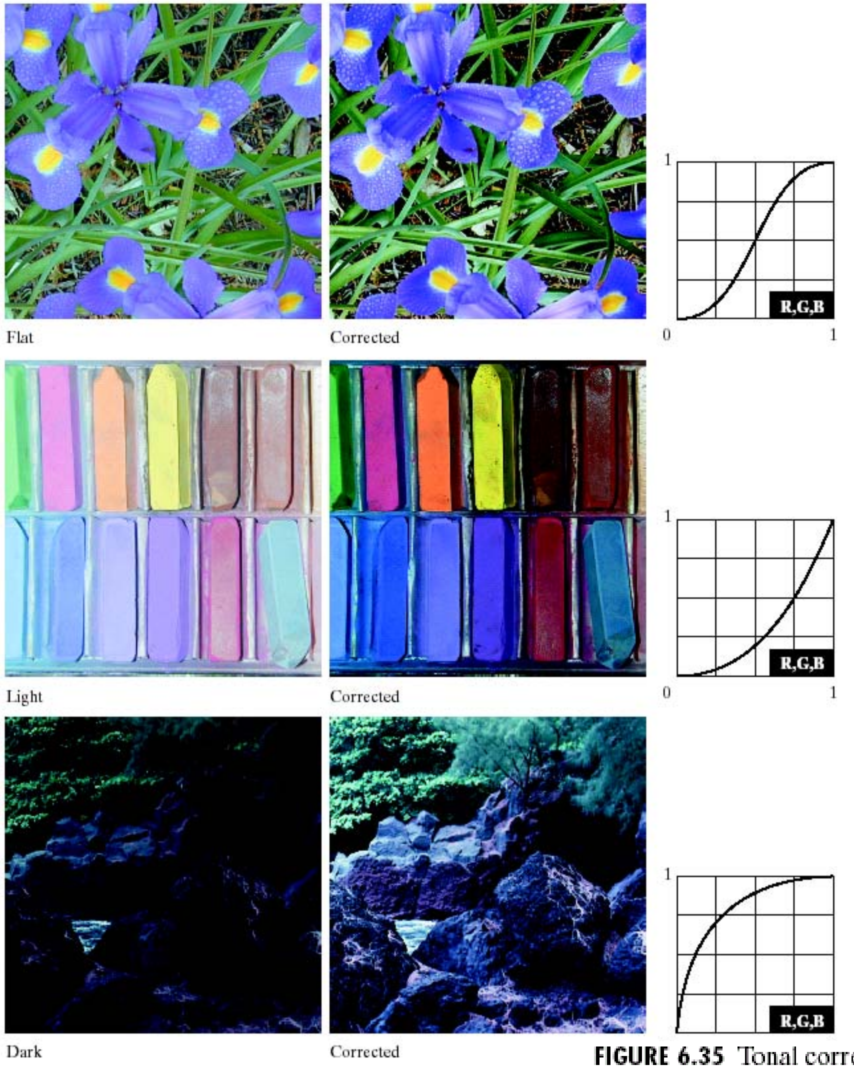


FIGURE 6.35 Tonal corrections for flat, light (high key), and dark (low key) color images. Adjusting the red, green, and blue components equally does not alter the image hues.

Color transformations-tone and color corrections

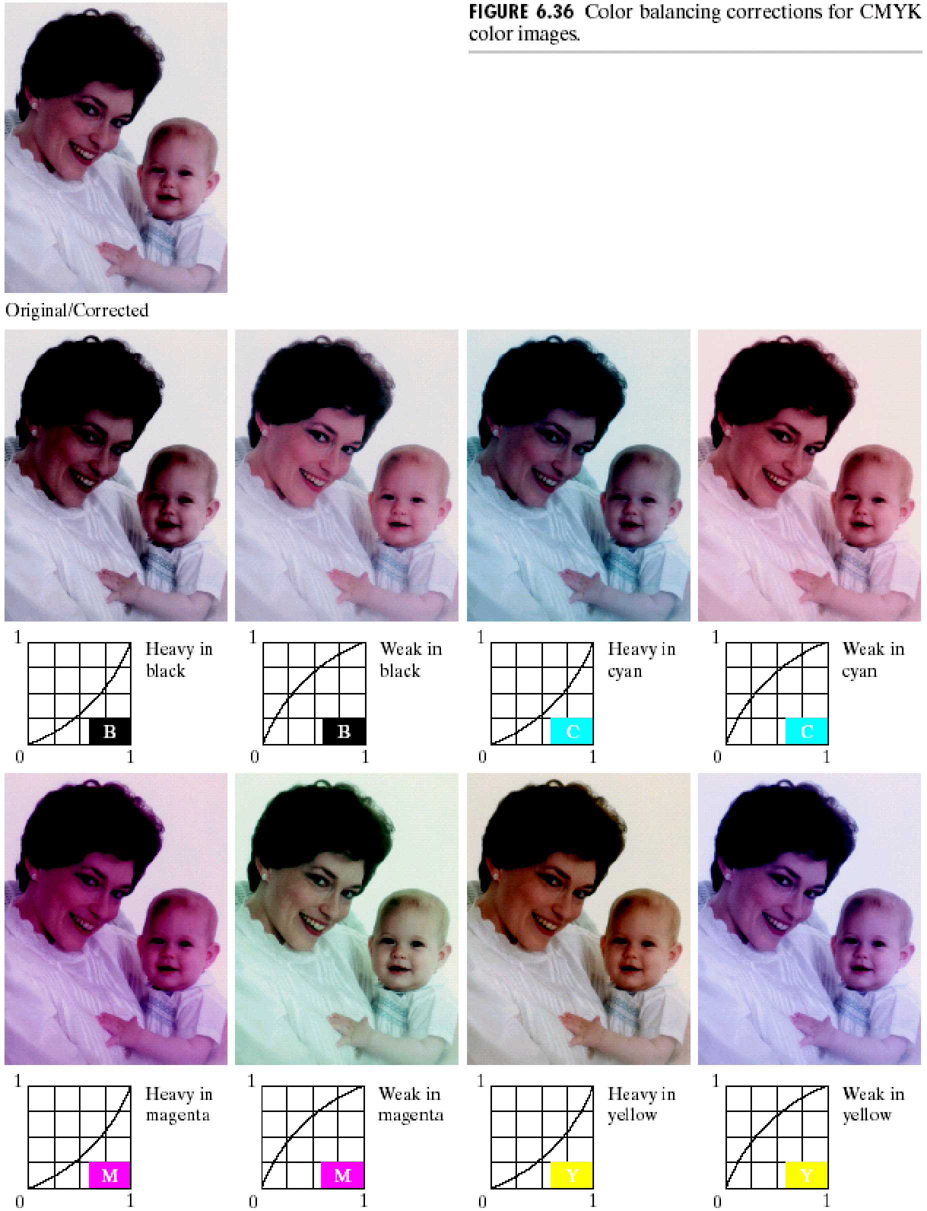
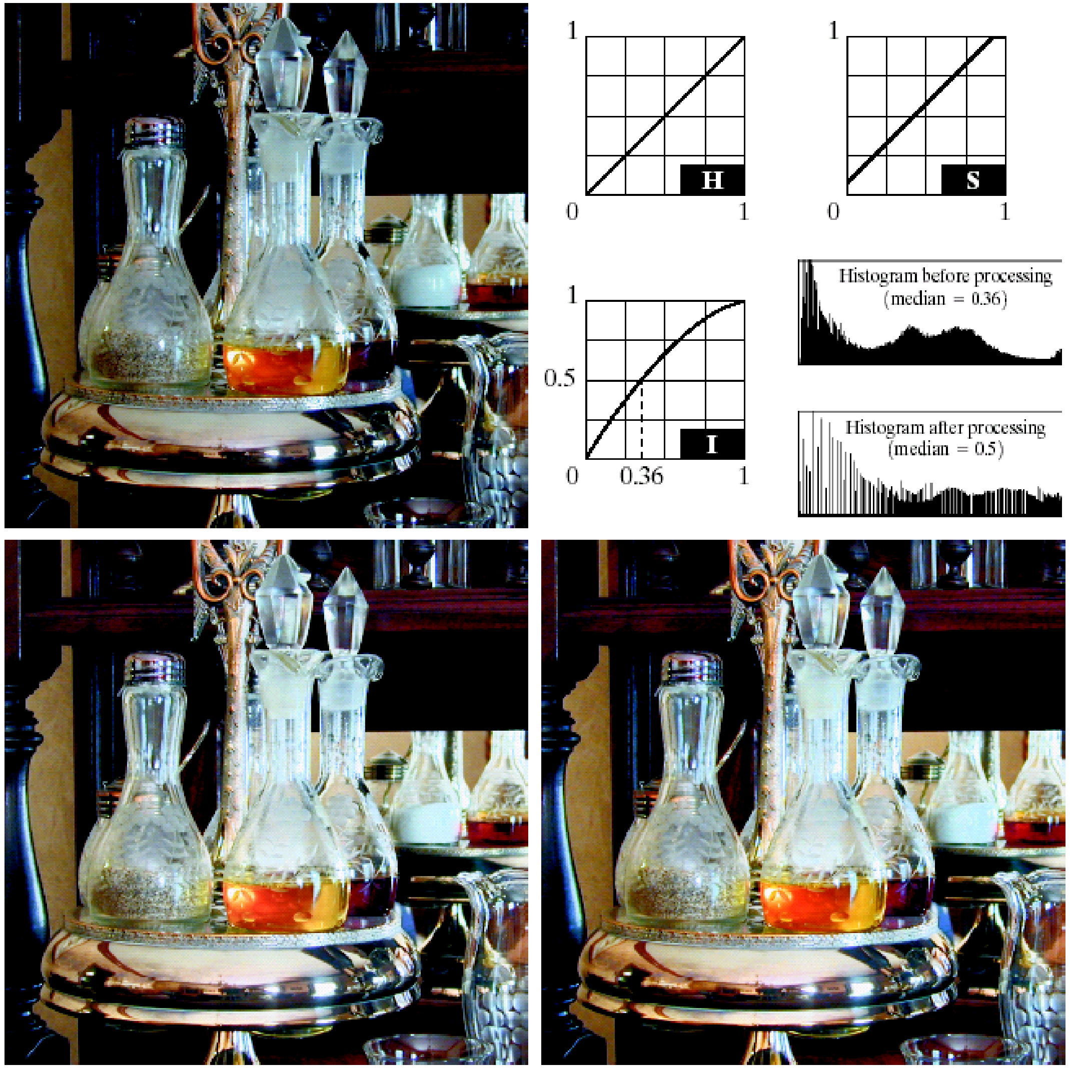


FIGURE 6.36 Color balancing corrections for CMYK color images.

Full color processing-
Color transformations-tone and color corrections



a b
c d

FIGURE 6.37 Histogram equalization (followed by saturation adjustment) in the HSI color space.

Full color processing-
Color smoothing and sharping

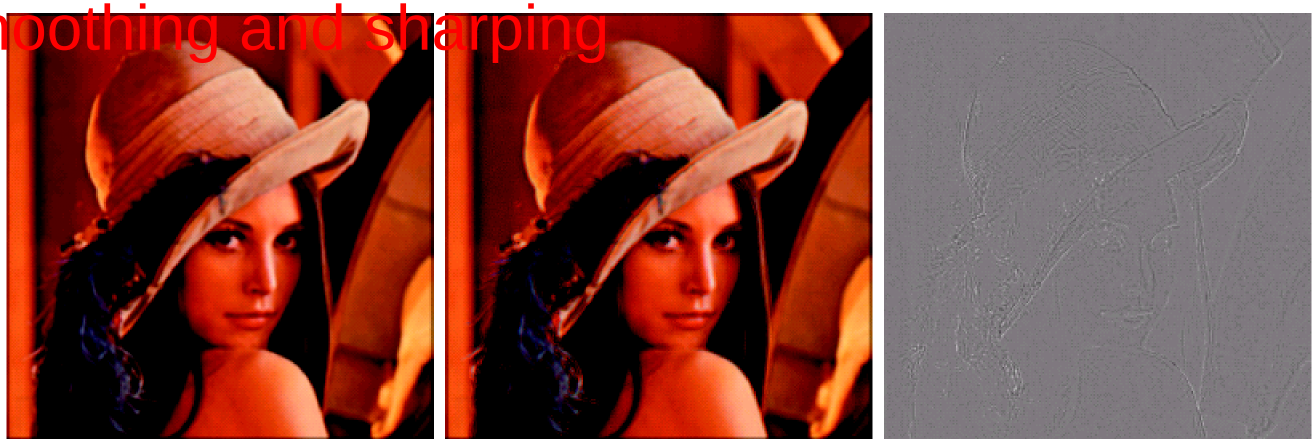
a b
c d

FIGURE 6.38 (a) RGB image. (b) Red component image. (c) Green component. (d) Blue component.

a b c

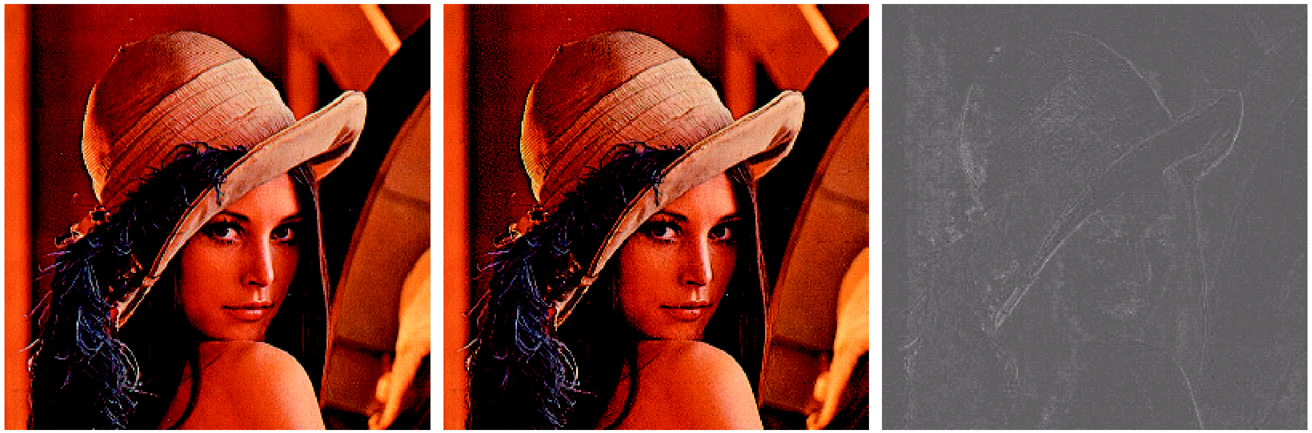
FIGURE 6.39 HSI components of the RGB color image in Fig. 6.38(a). (a) Hue. (b) Saturation. (c) Intensity.

Full color processing-
Color smoothing and sharping



a b c

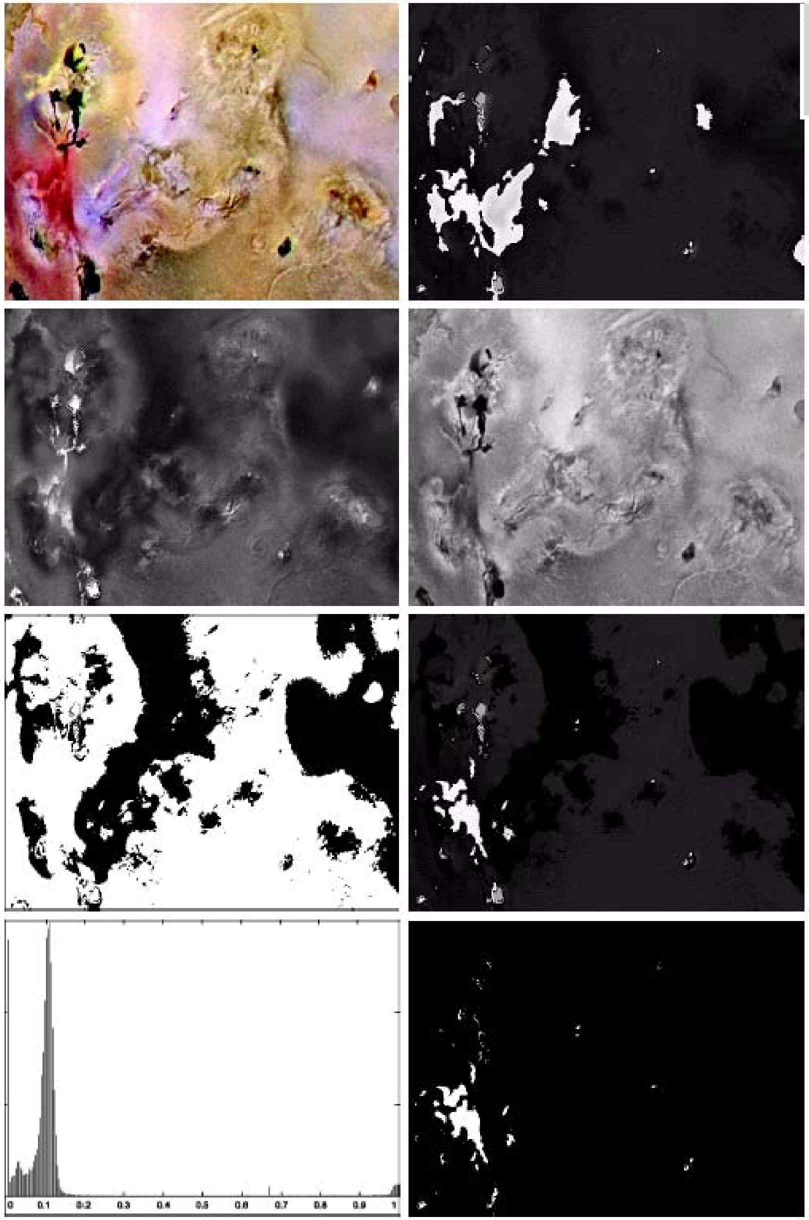
FIGURE 6.40 Image smoothing with a 5 × 5 averaging mask. (a) Result of processing each RGB component image. (b) Result of processing the intensity component of the HSI image and converting to RGB. (c) Difference between the two results.



a b c

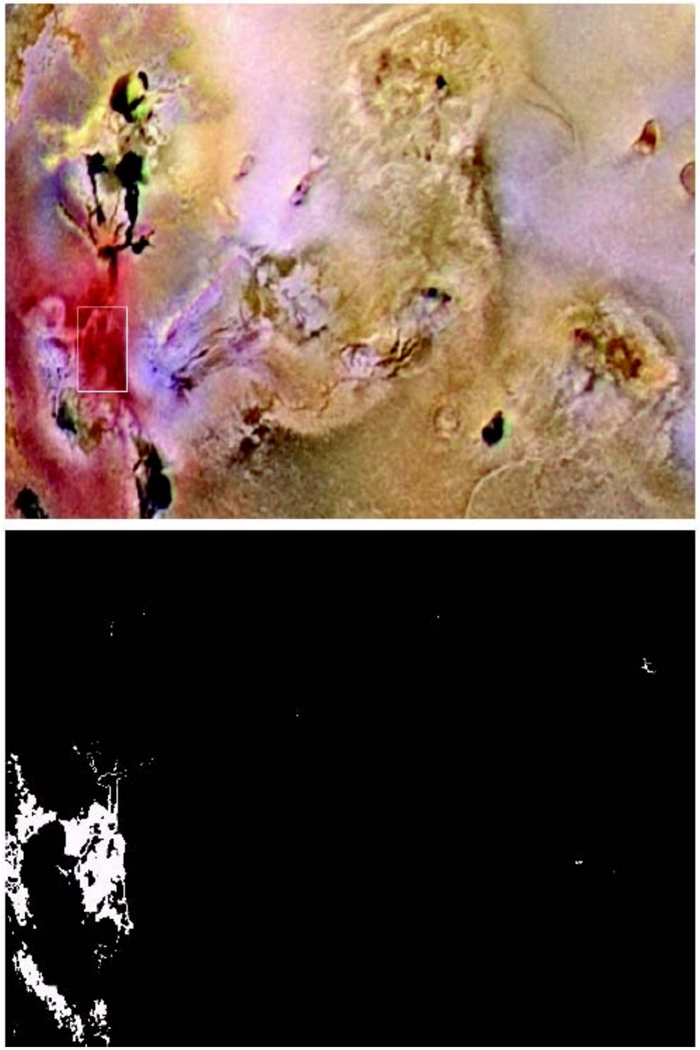
FIGURE 6.41 Image sharpening with the Laplacian. (a) Result of processing each RGB channel. (b) Result of processing the intensity component and converting to RGB. (c) Difference between the two results.

Image segmentation based on color



a b
c d
e f
g h

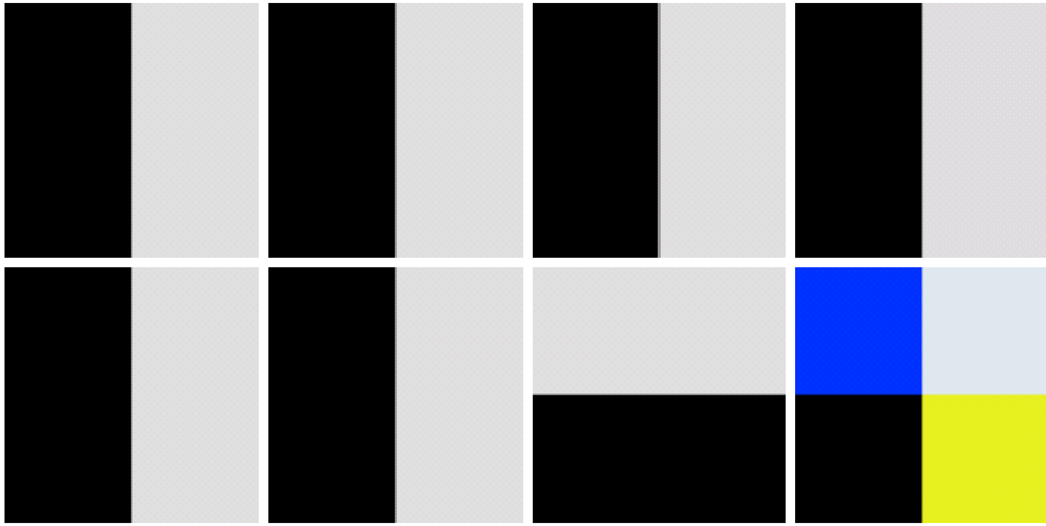
FIGURE 6.42 Image segmentation in HSI space. (a) Original. (b) Hue. (c) Saturation. (d) Intensity. (e) Binary saturation mask (black = 0). (f) Product of (b) and (e). (g) Histogram of (f). (h) Segmentation of red components in (a).



a
b

FIGURE 6.44 Segmentation in RGB space. (a) Original image with colors of interest shown enclosed by a rectangle. (b) Result of segmentation in RGB vector space. Compare with Fig. 6.42(h).

Color edge detection



a b c d
e f g h

FIGURE 6.45 (a)–(c) R, G, and B component images and (d) resulting RGB color image. (f)–(g) R, G, and B component images and (h) resulting RGB color image.

Color edge detection

a b
c d

FIGURE 6.46
(a) RGB image.
(b) Gradient computed in RGB color vector space.
(c) Gradients computed on a per-image basis and then added.
(d) Difference between (b) and (c).

a b c

FIGURE 6.47 Component gradient images of the color image in Fig. 6.46. (a) Red component, (b) green component, and (c) blue component. These three images were added and scaled to produce the image in Fig. 6.46(c).